



TECHPIOASSET — ENTERPRISE ITAM REDESIGN

UI / UX Design Pack

Roles, Journeys, IA, Mobile, Dashboards, Wireframes & Design System

Discipline extract from the 86-page master blueprint

Pack	UI / UX Design Pack
Audience	UI/UX designers, product designers, front-end engineers
Product	TechpioAsset — multi-tenant SaaS (Web + Mobile + REST API)
Version	2.0 — target-state redesign
Master document	TechpioAsset — Enterprise ITAM Redesign Blueprint (86 pp, BRD + FRD + Technical SRS)
Baseline	Monorepo at E:\TechpioAsset · live at https://piotask.com

SCOPE OF THIS PACK Everything a designer needs to lay out the redesign: what each of the 13 roles must see and do, end-to-end and onboarding/offboarding journeys, the status model that drives badges & filters, the 7-tab asset page, full web IA (module/screen/nav trees), the mobile-first app, all role dashboards, the reports each screen surfaces, and the complete UI design system with wireframes. Data schema, API internals and enforcement logic live in the Developer Pack.

Contents

- 1 Roles & What Each Role Needs (dashboards, KPIs, quick actions, mobile)
- 2 Business Flow & Screen Journeys (end-to-end, onboarding, offboarding)
- 3 Asset Status Model — for badges, filters & list views
- 4 Asset Detail — 7-Tab Hardware / Health Page
- 5 Information Architecture — Module, Screen & Navigation Trees (Web)
- 6 Mobile App Redesign (offline-first field companion)
- 7 Role-Based Dashboards (all 13 roles, with layouts)
- 8 Reports Catalogue (what each screen must surface)
- 9 UI Design System + Wireframes (tokens, components, screens)

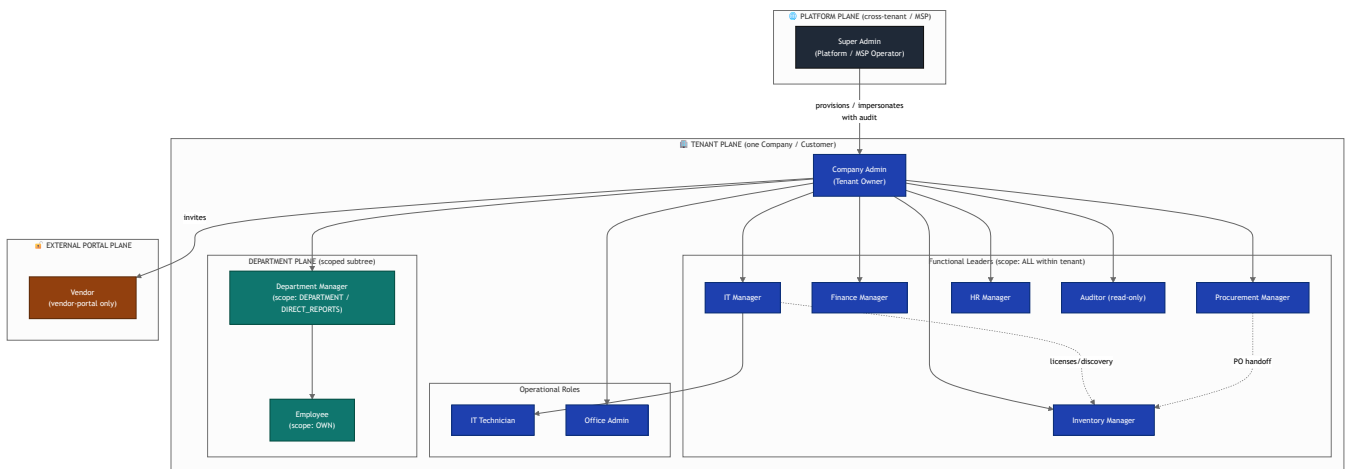
PART 1

Roles & What Each Role Needs (dashboards, KPIs, quick actions, mobile)

1. Complete Role Hierarchy

1.1 Governance layering (multi-tenant / MSP)

The platform enforces **three authority planes**. A principal is always anchored to exactly one plane; scopes narrow authority *within* a plane, they never cross a plane boundary upward.



1.2 How the planes work

Plane	Anchored roles	Tenant visibility	Purpose
Platform / MSP	Super Admin	Cross-tenant (all customers the operator manages)	Runs the SaaS/MSP fleet: provisions tenants, sets platform policy, aggregates fleet KPIs, supports customers via audited impersonation .
Tenant (Company)	Company Admin + all functional/operational roles	Single tenant only (hard row-level isolation)	One customer's ITAM world. Company Admin is the sovereign; functional leaders own their module vertically.
Department	Department Manager, Employee	Subtree of one tenant	Local, self-service and team-level asset ownership.
External Portal	Vendor	Only records explicitly shared with them (their POs, RMAs, contracts)	Zero access to the tenant app; a walled portal keyed to vendor_id .

1.3 Scope semantics (the four data scopes)

Every permission a role holds is evaluated against the **narrowest** scope granted for that permission group:

Scope	Row filter applied	Typical holder
ALL	All rows in the tenant (Super Admin: all rows in all managed tenants)	Company Admin, functional managers, Auditor (read)
DEPARTMENT	<code>record.department_id ∈ manager's managed departments</code>	Department Manager, Office Admin (site-scoped)

Scope	Row filter applied	Typical holder
DIRECT_REPORTS	<code>record.owner_id ∈ {users reporting to me}</code>	Department Manager (approvals)
OWN	<code>record.owner_id = me</code> OR <code>record.requester_id = me</code>	Employee, and every role for <i>their personal</i> assets

Composition rule: effective access = `permission_granted(role) AND scope_filter(role, permission_group)`. A permission with no scope resolves to `OWN` by default (fail-closed). Super Admin's `ALL` is additionally gated by an **MSP tenant-membership** check so an operator only sees tenants under their contract.

1.4 Legacy → canonical mapping (keep existing functionality)

Legacy role (8)	Canonical role	Notes
Admin	Company Admin	1:1
Manager	IT Manager	default; can be reassigned to Department Manager
Asset Manager	Inventory Manager	renamed, scope widened
Staff / User	Employee	scope <code>OWN</code>
Finance	Finance Manager	1:1
HR	HR Manager	1:1
Auditor	Auditor	read-only invariant enforced
Guest	Vendor or Employee (read)	routed by portal type

New roles introduced by the redesign: **Super Admin (MSP), IT Technician, Procurement Manager, Office Admin, Department Manager.**

2. Role Profiles

Each block: Responsibilities · Data scope · Dashboard · Key permissions · Primary workflows · Reports · Notifications · Mobile · Top KPIs.

2.1 Super Admin — Platform / MSP Operator (cross-tenant)

- **Responsibilities:** Provision/suspend tenants; set platform-wide policy, feature flags, SSO/SCIM, data-residency; manage license entitlements sold to tenants; fleet health; billing; audited customer support impersonation.
- **Data scope:** `ALL` across **all managed tenants** (MSP membership-gated).
- **Dashboard:** Fleet map (tenant count, active/suspended), aggregate asset count by tenant, per-tenant health & storage, license-pool consumption across fleet, discovery-agent connectivity %, security/incident feed, MRR/seat usage, impersonation-session log.
- **Key permissions:** `platform:tenant:*`, `platform:policy:*`, `platform:billing:*`, `platform:impersonate:execute`, `admin:*:read (cross-tenant)`, `discovery:connector:manage`.
- **Primary workflows:** Onboard tenant → seed roles/policies → assign entitlements; suspend/offboard with data-export; global patch/agent rollout; escalated support via impersonation.
- **Reports:** Fleet inventory rollup, cross-tenant license compliance, SLA/uptime, per-tenant consumption/billing, security posture.
- **Notifications:** Tenant provisioning events, agent/connector outages, quota breaches, security alerts, failed billing, impersonation start/stop.
- **Mobile:** Fleet health tile, critical-incident push, approve/suspend tenant, kill a runaway job.
- **Top KPIs:** Active tenants, fleet asset count, agent uptime %, cross-tenant compliance %, MRR, open Sev-1s.

Sees across	Can mutate	Cannot
All tenants (contracted)	Platform config, tenant lifecycle	Read tenant business data without an audited impersonation session

2.2 Company Admin — *Tenant Owner*

- **Responsibilities:** Full sovereignty of one tenant: users/roles, org tree, custom roles, integrations, categories/CMDB config, approval chains, branding, retention.
- **Data scope:** **ALL** within tenant.
- **Dashboard:** Total assets by status/category, license compliance gauge, open requests/approvals, maintenance due, spend YTD vs budget, user/seat count, warranty-expiry funnel, audit-exceptions tile.
- **Key permissions:** **admin:***, ***:*:*** (scope **ALL**, **tenant-bounded**) except platform plane; can create **custom roles** and edit approval workflows.
- **Primary workflows:** User/role provisioning; configure procurement approval chains; category & lifecycle setup; integration/SSO setup; period-end audit sign-off.
- **Reports:** Everything tenant-wide — inventory, compliance, spend, utilization, audit.
- **Notifications:** Approval escalations, compliance breaches, contract/warranty expiries, integration failures, unusual-activity alerts.
- **Mobile:** Approve anything, user lock/unlock, exec KPI board, config toggles.
- **Top KPIs:** Total asset value, license compliance %, request cycle time, spend vs budget, audit exceptions.

2.3 IT Manager

- **Responsibilities:** Owns hardware/software estate, discovery, licensing strategy, maintenance program, IT team & queue.
- **Data scope:** **ALL** (tenant) for IT asset/license/discovery/maintenance modules.
- **Dashboard:** Asset health by lifecycle, license compliance & true-up risk, discovery drift (unmanaged devices), maintenance SLA, technician workload, patch/EOL calendar, warranty pipeline.
- **Key permissions:** **assets:***, **license:***, **discovery:***, **maintenance:***, **inventory:read**, **requests:approve (IT)**, **reports:read**.
- **Primary workflows:** Triage IT requests → assign technician; license true-up; run discovery reconciliation; schedule maintenance/patch cycles; approve IT procurement specs.
- **Reports:** License compliance/position, hardware lifecycle & EOL, discovery reconciliation, maintenance SLA, technician throughput.
- **Notifications:** License overage, EOL/warranty, failed discovery sync, SLA breach, high-priority incident.
- **Mobile:** Approve IT requests, reassign tickets, scan asset, view technician map.
- **Top KPIs:** License compliance %, unmanaged-device %, MTTR, patch coverage %, warranty-at-risk count.

2.4 IT Technician

- **Responsibilities:** Execute assignments — image/deploy, check-in/out, repairs, moves, discovery agent installs.
- **Data scope:** **ALL** (read) for IT assets; **write** scoped to assigned tickets/assets (**DEPARTMENT** or assignment-based).
- **Dashboard:** My queue, assets awaiting deploy, check-outs due back, maintenance tasks today, low-stock consumables, recent scans.
- **Key permissions:** **assets:read**, **assets:update (assigned)**, **assets:checkin**, **assets:checkout**, **maintenance:execute**, **inventory:consume**, **discovery:agent:install**.
- **Primary workflows:** Assign/return asset to user; complete maintenance task; log repair; adjust stock on consume; onboard/offboard device.
- **Reports:** My task history, assets by location, maintenance-completed, check-out aging.

- **Notifications:** New assignment, overdue return, part arrived, escalation.
- **Mobile (heavy):** Barcode/QR/RFID scan, check-in/out, photo-attach condition, offline queue, 簽 signature capture, geo-stamp.
- **Top KPIs:** Tickets closed/day, on-time task %, overdue returns, mean handle time.

2.5 HR Manager

- **Responsibilities:** Owns joiner-mover-leaver (JML) asset events, personal issuance policy, offboarding asset recovery, acknowledgment/e-sign of custody.
- **Data scope:** **ALL** for **people/HR**; **read** on assets **assigned to employees**; no financial cost visibility beyond issuance.
- **Dashboard:** New hires needing kit, leavers with unreturned assets, acknowledgment-signature status, per-employee asset custody, policy exceptions.
- **Key permissions:** **people:***, **assets:read (assigned-to-people)**, **requests:create (onboarding)**, **requests:approve (people)**, **reports:read (custody)**.
- **Primary workflows:** Onboarding kit request → issuance; offboarding recovery checklist → clearance; custody acknowledgment e-sign; org-tree updates feeding scopes.
- **Reports:** Custody by employee/department, offboarding recovery, unreturned-on-exit, acknowledgment compliance.
- **Notifications:** New hire, termination date approaching, unreturned asset at exit, unsigned custody.
- **Mobile:** Approve onboarding kits, view employee custody, trigger offboarding, sign-off recovery.
- **Top KPIs:** Onboarding-kit SLA, offboarding recovery %, unreturned-at-exit count, acknowledgment %.

2.6 Finance Manager

- **Responsibilities:** Capitalization, depreciation, budgets, invoices/AP matching, TCO, cost centers, chargeback/showback.
- **Data scope:** **ALL** for **finance/invoices**; **read** on assets/procurement (cost view).
- **Dashboard:** Asset book value & depreciation curve, spend vs budget by cost center, open invoices/3-way-match exceptions, upcoming renewals cost, TCO by category, chargeback summary.
- **Key permissions:** **invoices:***, **finance:***, **procurement:read**, **assets:read (financials)**, **license:read (cost)**, **reports:***.
- **Primary workflows:** PO→invoice 3-way match & approve; run depreciation period-close; budget vs actual review; renewal cost forecasting; cost-center allocation.
- **Reports:** Depreciation schedule, TCO, budget variance, invoice-aging, renewal cost forecast, chargeback/showback.
- **Notifications:** Invoice-match exception, budget overrun, renewal cost due, month-end close tasks.
- **Mobile:** Approve invoices, view spend tile, budget alerts.
- **Top KPIs:** Spend vs budget, net book value, TCO/asset, invoice-match exception rate, renewal cost at risk.

2.7 Procurement Manager

- **Responsibilities:** Requisitions→POs, vendor/catalog management, sourcing, contract terms, receiving oversight.
- **Data scope:** **ALL** for **procurement**; **read** /manage on **vendor-portal** invitations; **read** inventory & assets.
- **Dashboard:** Open requisitions by stage, POs pending/receiving, vendor SLA/lead-time, catalog price trends, contract-renewal pipeline, spend by vendor.
- **Key permissions:** **procurement:***, **vendor-portal:invite**, **vendor-portal:manage**, **inventory:read**, **invoices:read**, **assets:create (on receipt)**.

- **Primary workflows:** Approve requisition → raise PO → dispatch to vendor portal → track → confirm receipt handoff to Inventory.
- **Reports:** PO cycle time, vendor performance/scorecard, price variance, contract-renewal calendar, spend-by-category.
- **Notifications:** New requisition, PO acknowledged/shipped by vendor, late delivery, contract renewal, price change.
- **Mobile:** Approve requisition/PO, vendor status, receive-confirm.
- **Top KPIs:** PO cycle time, on-time delivery %, price variance, vendor SLA %, contract renewals at risk.

2.8 Inventory Manager

- **Responsibilities:** Stockroom/warehouse, stock levels, GRN/receiving, transfers, cycle counts, consumables, bin/location.
- **Data scope:** **ALL** for **inventory** (optionally **DEPARTMENT** /site-scoped in multi-site tenants).
- **Dashboard:** Stock on hand vs reorder, incoming receipts, low-stock/out-of-stock, transfers in-flight, cycle-count variance, shrinkage, aging stock.
- **Key permissions:** **inventory:***, **assets:create (from stock)**, **assets:transfer**, **procurement:read (POs to receive)**, **reports:read**.
- **Primary workflows:** Goods receipt (GRN) against PO → stock-in → convert to trackable asset → transfer/issue; cycle count & reconcile; reorder trigger.
- **Reports:** Stock valuation, reorder report, cycle-count variance, movement/turns, shrinkage.
- **Notifications:** Reorder point hit, incoming shipment, count variance, expiry (consumables), transfer received.
- **Mobile (heavy):** Scan receive, bin move, cycle-count mode, stock lookup, adjust qty.
- **Top KPIs:** Stock accuracy %, stockout rate, inventory turns, cycle-count variance, shrinkage %.

2.9 Office Admin

- **Responsibilities:** Site/facility assets (furniture, peripherals, access cards, keys), local issuance, front-desk requests — one or more sites.
- **Data scope:** **DEPARTMENT** (site-scoped) for assets/inventory; **read** broader.
- **Dashboard:** Site asset roster, local check-outs, pending local requests, low office-supply stock, visitor/loaner equipment, upcoming returns.
- **Key permissions:** **assets:read (site)**, **assets:checkin/out (site)**, **inventory:read/consume (site)**, **requests:create**, **requests:approve (site, low-value)**.
- **Primary workflows:** Local asset issuance/return; loaner desk; office-supply replenish request; new-desk setup coordination.
- **Reports:** Site inventory, local check-out aging, supply consumption.
- **Notifications:** Local request, overdue loaner, low supplies, new-hire desk setup.
- **Mobile:** Scan/check-out at desk, approve low-value site request, supply reorder.
- **Top KPIs:** Local request SLA, loaner return %, site asset accuracy, supply stockouts.

2.10 Department Manager

- **Responsibilities:** Approves team asset/software requests; owns department budget consumption; visibility into team custody.
- **Data scope:** **DEPARTMENT** + **DIRECT_REPORTS** (approvals over reports; view over department).
- **Dashboard:** Team assets & value, pending approvals, department spend vs allocation, licenses assigned to team, unreturned by leavers on team, requests in flight.

- **Key permissions:** `requests:approve (DIRECT_REPORTS)`, `assets:read (DEPARTMENT)`, `license:read (DEPARTMENT)`, `people:read (DIRECT_REPORTS)`, `reports:read (DEPARTMENT)`.
- **Primary workflows:** First-level approve/reject team requests; endorse hardware refresh; confirm team custody in audits; department budget check.
- **Reports:** Team custody, department spend, license utilization (team), request approvals log.
- **Notifications:** Approval pending, budget threshold, team member offboarding, audit confirmation due.
- **Mobile:** One-tap approve/reject, team asset list, budget tile.
- **Top KPIs:** Approval turnaround, department spend vs budget, team license utilization, custody-confirmed %.

2.11 Employee

- **Responsibilities:** Requests assets/software; acknowledges custody; reports issues; returns assets.
- **Data scope:** `OWN`.
- **Dashboard:** My assets, my open requests, my licenses, return-due items, my tickets, acknowledgment tasks.
- **Key permissions:** `assets:read (OWN)`, `requests:create`, `requests:read (OWN)`, `maintenance:report (OWN)`, `license:read (OWN)`, `people:read (self)`.
- **Primary workflows:** Raise request → track → receive → acknowledge; report fault; return on exit/refresh.
- **Reports:** My assets/custody statement (self-service).
- **Notifications:** Request status, asset assigned, return due, acknowledgment required, maintenance scheduled for my device.
- **Mobile (heavy):** Self-service catalog request, scan-to-report-issue, e-sign custody, view my kit, check-in return.
- **Top KPIs:** My open requests, items due for return, acknowledgment status.

2.12 Auditor — *read-only invariant*

- **Responsibilities:** Independent assurance — inventory accuracy, license compliance, control effectiveness, evidence collection; **no mutation anywhere**.
- **Data scope:** `ALL read-only` (tenant); Super-Auditor variant may be cross-tenant for MSP compliance.
- **Dashboard:** Compliance heatmap, audit-exception register (read), evidence completeness, control-test status, change/access log viewer, discovery-vs-record variance.
- **Key permissions:** `*:*:read` only; `audit:evidence:export`; **explicitly denied** every `create/update/delete/approve` (enforced by a hard invariant, see §3).
- **Primary workflows:** Sample assets → verify existence → log finding (in a separate immutable findings store, not the asset records) → export evidence pack.
- **Reports:** Full read of all reports + audit-trail, access-review, license-compliance, reconciliation, SoD-conflict.
- **Notifications:** New audit period, control-test due, evidence-gap alerts (informational only).
- **Mobile:** Read dashboards, scan-to-verify (records verification event in findings log only), export.
- **Top KPIs:** Inventory accuracy %, open findings, evidence completeness %, compliance score.

2.13 Vendor — *External Portal (walled)*

- **Responsibilities:** External supplier — acknowledges POs, updates shipment/delivery, submits invoices, manages RMAs/warranty claims, maintains catalog entries.
- **Data scope:** **Only records linked to** `vendor_id` — their POs, contracts, RMAs, invoices. No tenant app access.
- **Dashboard (portal):** POs to acknowledge, shipments in progress, open RMAs, invoice status, contract/SLA summary, catalog items.
- **Key permissions:** `vendor-portal:po:read`, `vendor-portal:po:acknowledge`, `vendor-portal:shipment:update`, `vendor-portal:invoice:submit`, `vendor-portal:rma:manage`, `vendor-portal:catalog:update`.
- **Primary workflows:** Acknowledge PO → update shipping/tracking → submit invoice → handle RMA/warranty.

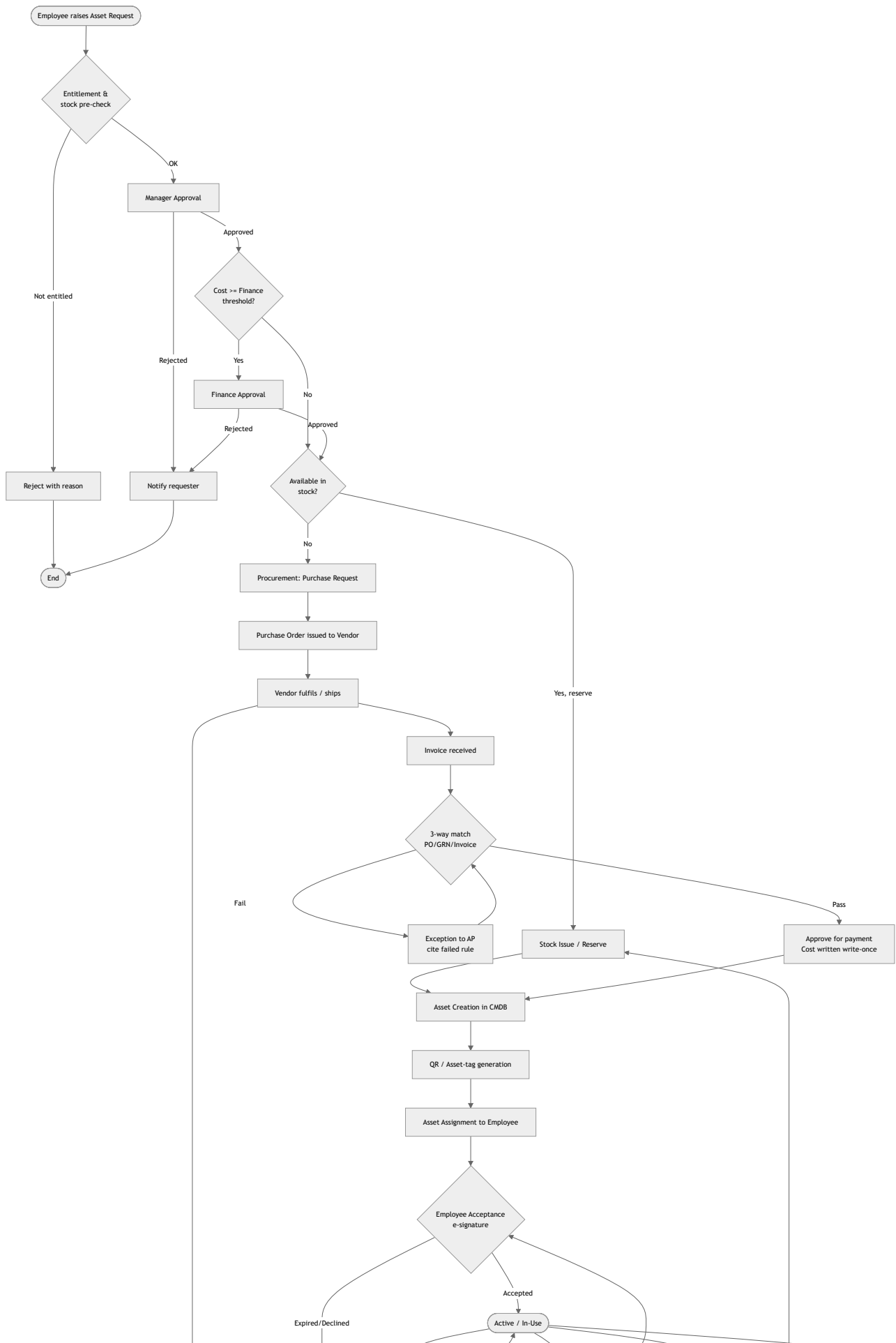
- **Reports (portal):** My PO history, delivery performance (self), invoice/payment status.
 - **Notifications:** New PO, delivery reminder, RMA opened, invoice approved/rejected, contract renewal.
 - **Mobile (portal):** Acknowledge PO, update tracking, upload invoice/shipping docs.
 - **Top KPIs:** PO acknowledgment time, on-time delivery %, invoice acceptance %, open RMAs.
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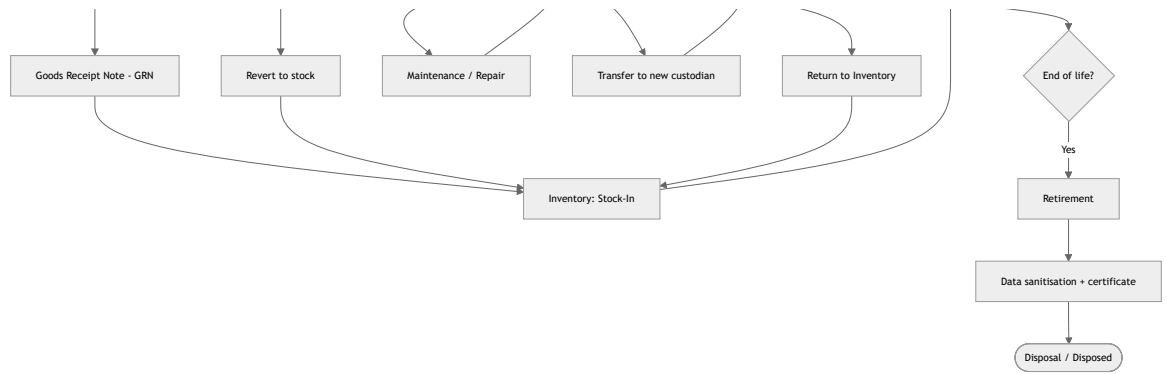
PART 2

Business Flow & Screen Journeys (end-to-end, onboarding, offboarding)

Part B — Complete End-to-End Business Flow

B.1 Mermaid Flowchart





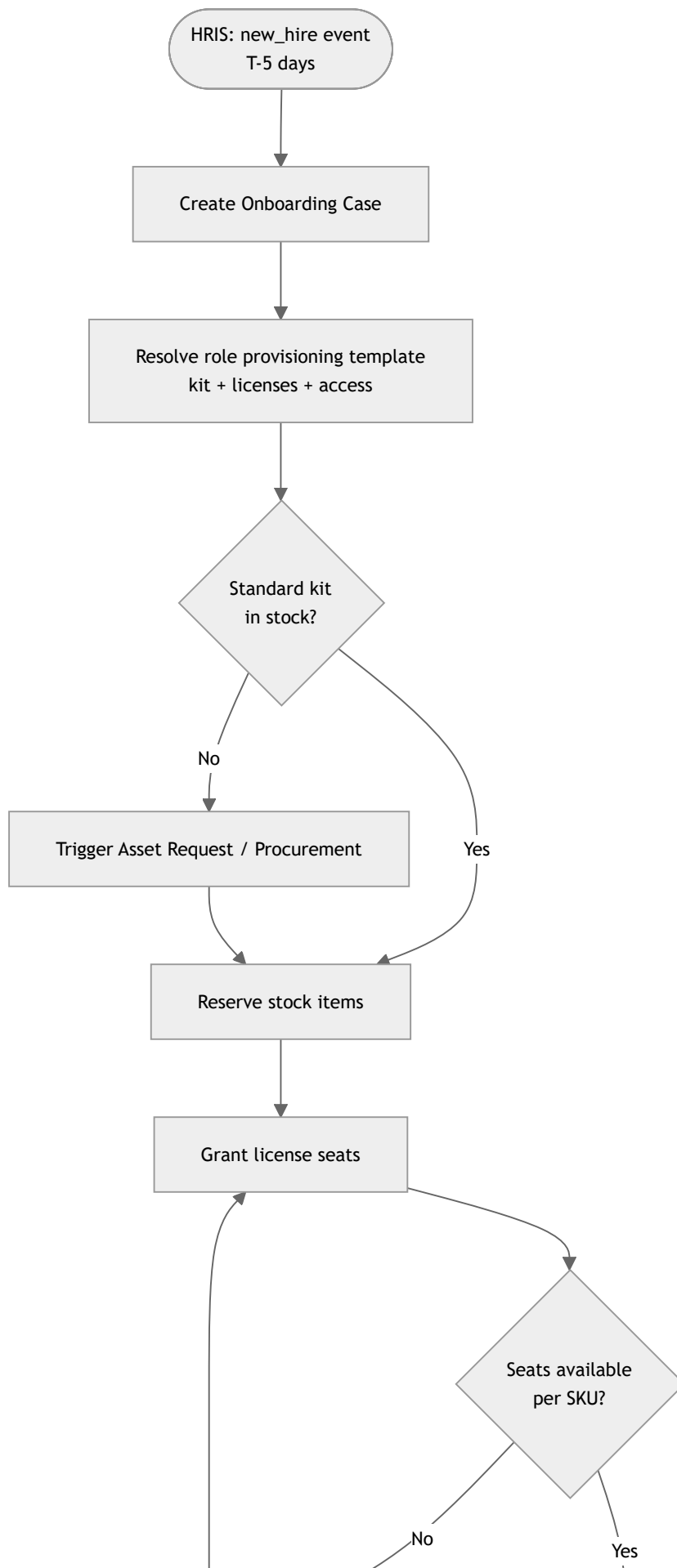
B.2 Numbered Step Table

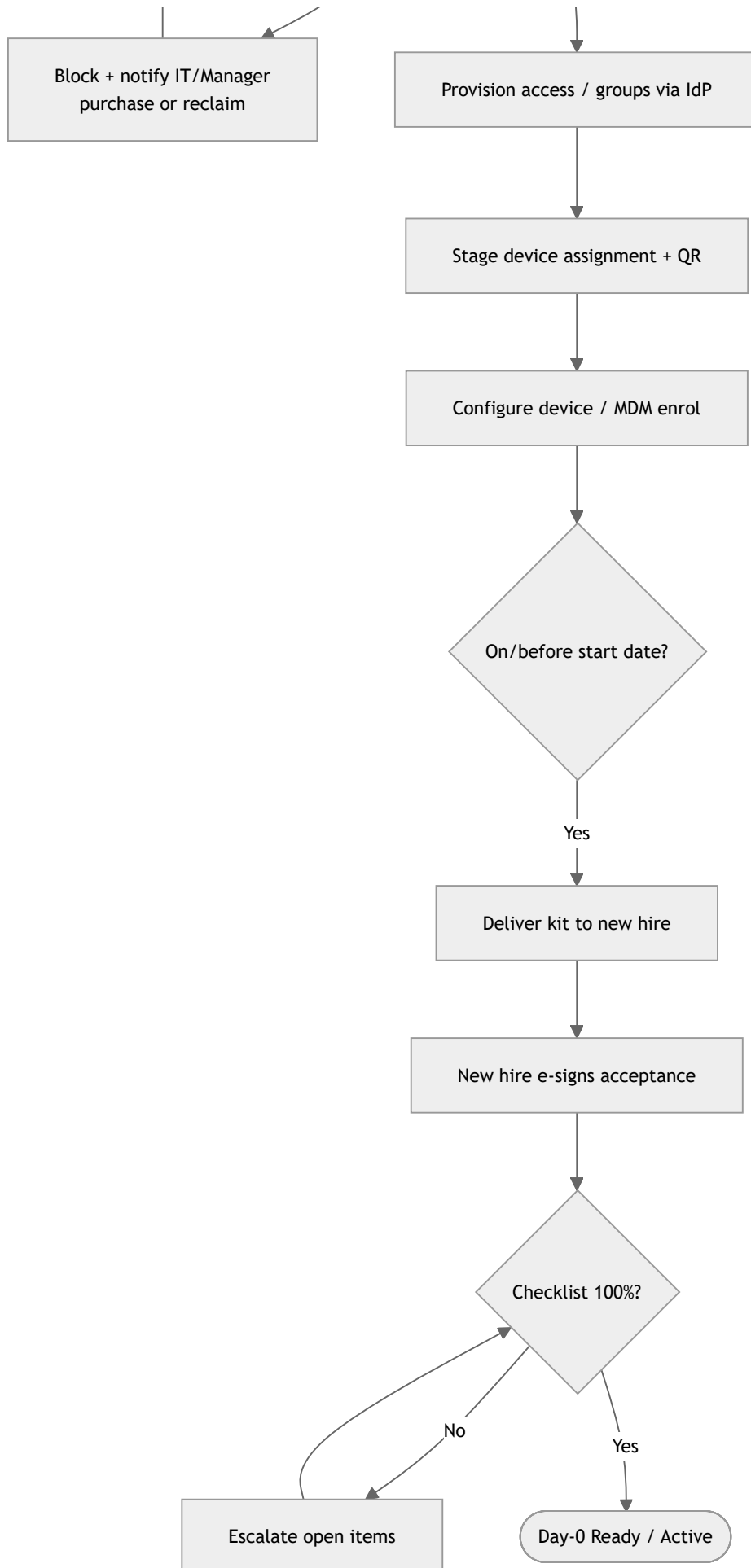
#	Step	Actor / Role	Entry Data	Validations (BR)	System Actions	Notifications	Resulting State
1	Asset Request	Employee	Catalogue item/kit, qty, justification, need-by date, cost centre	Scope & entitlement (BR-01/02); stock pre-check (BR-22)	Create request; estimate cost; determine standard vs non-standard	Requester ack; manager queued	Requested
2	Manager Approval	Line Manager	Approve/reject + comment	SoD (BR-04); scope (BR-02); SLA (BR-06)	Advance workflow; log decision	Requester; Finance if threshold	Manager-Approved
3	Finance Approval (conditional)	Finance	Approve/reject, budget check	Triggered if cost ≥ threshold (BR-05); SoD (BR-04)	Insert & resolve finance step	Requester; Procurement	Finance-Approved
4	Fulfilment Decision	IT Asset Mgr / system	Stock levels	Reservation lock (BR-22)	Route to stock-issue or procurement	Procurement / warehouse	Approved-For-Fulfilment
5	Purchase Request	Procurement	Vendor, SKU, qty, price	Scope; budget reference	Generate PR; select vendor	Approver of PR (if configured)	PR-Raised
6	Purchase Order	Procurement	PR ref, agreed price, terms	PR approved; vendor active	Issue PO; send to vendor	Vendor; Finance (I)	PO-Issued
7	Vendor Fulfilment	Vendor	Ship notice, tracking	–	Record ASN; await goods & invoice	IT Asset Mgr	Awaiting-Receipt
8	Goods Receipt (GRN)	IT Asset Mgr / Warehouse	Received qty, serials, condition	Qty ≤ PO qty; serial uniqueness (BR-20); GRN required (BR-10)	Create GRN; capture serials	Finance; Procurement	Received
9	Stock-In	Inventory / system	GRN lines	Location valid; scope	Increment inventory; create stock items	IT Asset Mgr	In-Stock
10	Invoice Receipt	Vendor / AP	Invoice lines, tax, currency	Duplicate check (BR-09)	Ingest invoice; parse lines	AP	Invoice-Received

#	Step	Actor / Role	Entry Data	Validations (BR)	System Actions	Notifications	Resulting State
11	3-Way Match	System (deterministic)	PO, GRN, Invoice	Match qty/price/tax within tolerance (BR-08); currency/FX (BR-23)	Auto-approve or raise exception citing rule	AP on exception	Matched / Exception
12	Cost Write (write-once)	System / Finance	Verified cost	Immutable cost (BR-07)	Append cost to ledger; schedule payment	Finance	Cost-Finalised
13	Asset Creation	IT Asset Mgr / system	Serial, model, cost ref, warranty/AMC	Uniqueness (BR-20); coverage dates (BR-14)	Create CMDB asset; link cost/PO/GRN	–	Registered
14	QR / Tag Generation	System	Asset ID	Unique QR payload (BR-20)	Generate QR/barcode + printable label	IT Asset Mgr	Tagged
15	Assignment	IT Asset Mgr	Assignee, location, custody terms	Scope; seat grant for bundled licenses (BR-11)	Assign; grant license seats; start acceptance timer	Assignee	Assigned
16	Acceptance (e-sign)	Employee	Signature, condition ack	Acceptance required (BR-15); expiry (BR-15)	Record signature; bind custody	Assignee; IT (I)	Active/In-Use
17	Maintenance / Repair	IT Asset Mgr / Vendor	Ticket, warranty status	Out-of-warranty needs approval (BR-14)	Log service event; AMC claim; loaner if needed	Custodian	In-Maintenance → Active
18	Transfer	IT Asset Mgr	New custodian/location	Scope; re-acceptance required (BR-15)	Reassign; reclaim/re-grant seats (BR-12)	Old & new custodians	In-Transfer → Active
19	Return	Employee / IT	Return reason, condition	Condition check	Unassign; reclaim seats (BR-12); inspect	IT Asset Mgr	Returned → In-Stock
20	Retirement	IT Asset Mgr	EoL reason, approval	Legal state transition (BR-21); approval	Flag EoL; remove from active pool	Security/Compliance	Retired
21	Data Sanitisation	Security / Vendor	Wipe/destroy method	Certificate required (BR-18)	Attach sanitisation certificate	Compliance	Sanitised
22	Disposal	IT Asset Mgr / Vendor	Disposal method, certificate	Certificate required (BR-18); immutability (BR-16)	Record disposal cert; finalise ledger	Finance; Compliance	Disposed

Part C — Employee Onboarding Workflow

C.1 Mermaid Flowchart



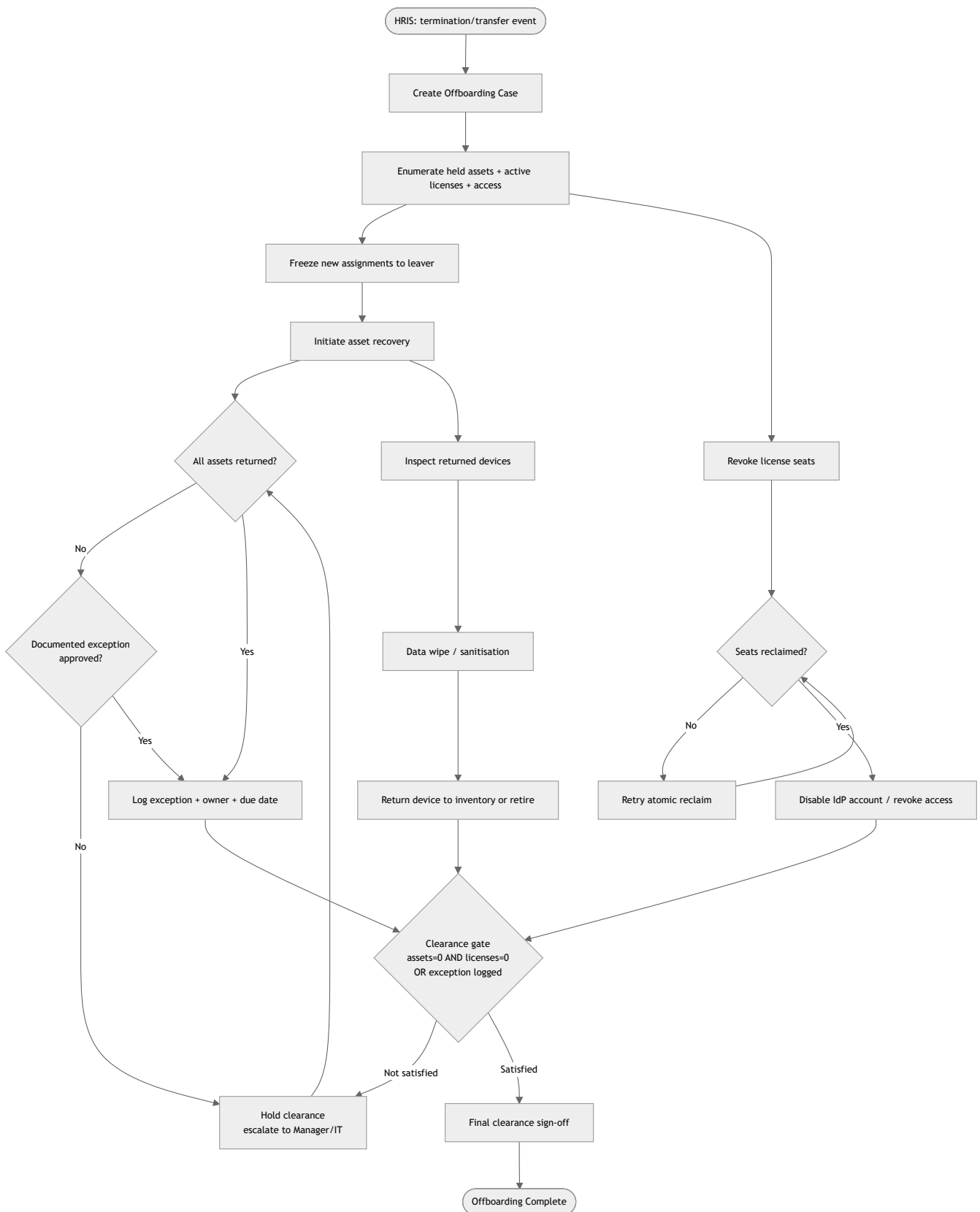


C.2 Onboarding Step Table

#	Step	Trigger / Actor	Entry Data	Validations (BR)	System Actions	Notifications	State
1	New-hire event	HRIS (system)	Employee, role, org unit, start date, location	Tenant scope (BR-03)	Create onboarding case	HR; IT Asset Mgr	Onboarding-Initiated
2	Template resolution	System	Role → provisioning template	Template exists for role	Expand kit + license + access bundle	IT	Template-Applied
3	Kit sourcing	System / IT	Kit items	Stock check (BR-22); reservation	Reserve stock or raise procurement (Part B flow)	Procurement if needed	Kit-Reserved / Kit-On-Order
4	License grant	System / IT Asset Mgr	SKUs, seat count	Seat ceiling (BR-11); scope	Consume seats atomically; ledger update	IT; Finance (I)	Licenses-Granted
5	Access provisioning	IT / IdP integration	Groups, roles, apps	Least-privilege; SoD	Push access to IdP/SaaS	Security (I)	Access-Provisioned
6	Device staging	IT Asset Mgr	Asset, QR tag	Uniqueness (BR-20)	Assign asset; generate QR; MDM enrol	IT	Device-Staged
7	Acceptance	New hire	E-signature	Acceptance gate (BR-15)	Bind custody; start warranty clock	New hire	Accepted / Active
8	Completion	System	Checklist status	All items closed	Close case; audit stamp (BR-16)	HR; Manager	Day-0 Ready

Part D — Employee Offboarding Workflow

D.1 Mermaid Flowchart



D.2 Offboarding Step Table

#	Step	Trigger / Actor	Entry Data	Validations (BR)	System Actions	Notifications	State
1	Termination event	HRIS (system)	Employee, last day, type (exit/transfer)	Tenant scope (BR-03)	Create offboarding case	HR; IT; Manager	Offboarding-Initiated
2	Inventory of holdings	System	Assigned assets, active licenses, access	Scope	Enumerate custody chain + seat ledger	IT Asset Mgr	Holdings-Listed
3	Freeze	System	Leaver identity	–	Block new assignments/grants to leaver	IT	Frozen
4	Asset recovery	IT / Employee	Return logistics	Condition capture	Generate return tasks; track each asset	Employee; Manager	Recovery-In-Progress
5	Device inspection	IT Asset Mgr	Returned condition	–	Log condition; flag damage/loss	Finance (I) if loss	Inspected
6	Data wipe / sanitisation	IT / Security	Wipe method	Certificate (BR-18)	Attach sanitisation record	Security	Sanitised
7	Return / retire	IT Asset Mgr	Disposition	State machine (BR-21)	Stock-in or route to retirement (Part B)	Inventory	In-Stock / Retired
8	License revocation	System / IT	SKUs held	Atomic reclaim (BR-12)	Reclaim seats to pool	Finance (I)	Seats-Reclaimed
9	Access disable	IT / IdP	Accounts, groups	Least-privilege exit	Disable/expire IdP + SaaS access	Security	Access-Revoked
10	Clearance gate	System	Assets outstanding, active licenses	BR-19: complete only if assets=0 AND licenses=0, else approved exception logged	Evaluate gate; block or allow	Manager; IT; Compliance	Clearance-Pending / Cleared
11	Exception (if needed)	Authorised approver	Reason, owner, due date	Exception must be approved & logged (BR-19)	Record documented exception; keep case open with follow-up	Compliance; Manager	Exception-Logged
12	Final sign-off	IT Asset Mgr / Manager	Confirmation	Immutable stamp (BR-16); GDPR pseudonymisation scheduled (BR-17)	Close case; write audit record	HR; Manager	Offboarding-Complete

Gating rule (BR-19) restated: Offboarding **cannot** transition to **Offboarding-Complete** while any asset remains in the leaver's custody or any paid license seat remains active for the leaver — **unless** an authorised role has approved and logged a documented exception (with owner and remediation due date), which keeps the case open for follow-up rather than silently closing it.

End of blueprint section. All diagrams are provided as Mermaid source within fenced blocks; all business rules are cross-referenced (BR-xx) throughout the flows and step tables for traceability into design and test coverage.

PART 3

Asset Status Model — for badges, filters & list views

1. Separated Status Model (the critical redesign)

1.1 Why the single 18-value enum must die

The current `Asset.status` field conflates five *orthogonal* facts into one column: where the asset is in its life, whether it can be given to someone, its physical shape, who owns it, and where it physically sits. Any real query ("show me all *leased* laptops that are *deployed* but in *poor* condition and physically in *Warehouse-BLR*") is impossible against one enum, and every new business case forces yet another enum value (that is how you get to 18).

The redesign models **five independent dimensions**, each a small closed enum or a relation, so state is a *tuple* not a single label.

1.2 The five dimensions

Dimension	Field on <code>Asset</code>	Kind	Values / Target	Semantics
Lifecycle	<code>lifecycleState</code>	enum <code>LifecycleState</code>	PLANNED, IN_PROCUREMENT, IN_STOCK, DEPLOYED, IN_MAINTENANCE, RETIRED, DISPOSED	Where the asset sits in its cradle-to-grave journey. Monotonic-ish; drives depreciation & CMDB visibility.
Availability	<code>availabilityState</code>	enum <code>AvailabilityState</code>	AVAILABLE, RESERVED, ASSIGNED, IN_TRANSIT, IN_REPAIR, LOST	Can it be handed out <i>right now</i> ? Drives the request/reservation engine.
Condition	<code>conditionGrade</code>	enum <code>ConditionGrade</code>	NEW, GOOD, FAIR, POOR, DAMAGED, END_OF_LIFE	Physical/functional health. Backed by an append-only <code>ConditionLog</code> history.
Ownership	<code>ownershipType</code>	enum <code>OwnershipType</code>	OWNED, LEASED, RENTED, BYOD, LOANER	Financial/legal ownership. Drives depreciation vs. lease-liability accounting.
Location	<code>currentLocationId</code> or <code>assignedToUserId</code>	FK (nullable)	→ <code>Location</code> / → <code>User</code>	Physical placement. Structured hierarchy (Site→Building→Floor→Room→Warehouse→Bin) or "with-user".

These are **independent**: `lifecycleState` does not determine `conditionGrade`, ownership is orthogonal to availability, etc. Guard *illegal* combinations with lightweight rules (see §1.4) rather than by re-merging into one enum.

```
enum LifecycleState { PLANNED IN_PROCUREMENT IN_STOCK DEPLOYED IN_MAINTENANCE RETIRED DISPOSED }
enum AvailabilityState{ AVAILABLE RESERVED ASSIGNED IN_TRANSIT IN_REPAIR LOST }
enum ConditionGrade { NEW GOOD FAIR POOR DAMAGED END_OF_LIFE }
enum OwnershipType { OWNED LEASED RENTED BYOD LOANER }
```

1.3 Where each dimension physically lives

- **Lifecycle / Availability / Condition / Ownership** → four scalar enum columns on `Asset` (fast to filter, cheap to index). Each column is `NOT NULL` with a sensible default.
- **Condition history** → separate `ConditionLog` table (append-only) so the *current* `conditionGrade` on `Asset` is a denormalized cache of the latest log row.
- **Lifecycle history** → `AssetLifecycleEvent` (append-only) records every transition with actor, reason, timestamp — the audit spine of the asset.
- **Location** → a normalized `Location` hierarchy table (self-referencing) referenced by `Asset.currentLocationId`. "With-user" is expressed by `Asset.assignedToUserId` being non-null; a `CHECK` ensures an asset is *either* at a location *or* with a user, never ambiguously both for its "current" placement (the assignment record carries the authoritative link).

1.4 Legal-combination guardrails (not a merged enum)

Enforced in the service layer + a few DB `CHECK`s / partial indexes:

- `availabilityState = ASSIGNED` ⇒ an active `AssetAssignment` row must exist (and `assignedToUserId` set).
- `lifecycleState IN (RETIRED, DISPOSED)` ⇒ `availabilityState` must be `LOST` or the asset unassigned; blocks new assignments.
- `lifecycleState = IN_MAINTENANCE` typically pairs with `availabilityState = IN_REPAIR` — warned, not hard-blocked (an asset can be flagged for maintenance while still physically assigned).
- `conditionGrade = END_OF_LIFE` ⇒ recommend transition to `RETIRED` (soft rule surfaced in UI).

1.5 Mapping the OLD 18 statuses → new dimensions

#	Old single status	LifecycleState	AvailabilityState	ConditionGrade	OwnershipType	Location note
1	PLANNED	PLANNED	AVAILABLE	NEW	(inherit)	—
2	ON_ORDER / ORDERED	IN_PROCUREMENT	RESERVED	NEW	(inherit)	vendor
3	IN_PROCUREMENT	IN_PROCUREMENT	RESERVED	NEW	(inherit)	vendor
4	RECEIVED	IN_STOCK	AVAILABLE	NEW	OWNED	warehouse/bin
5	IN_STOCK / AVAILABLE	IN_STOCK	AVAILABLE	GOOD	(inherit)	warehouse/bin
6	RESERVED	IN_STOCK	RESERVED	GOOD	(inherit)	warehouse/bin
7	ALLOCATED	IN_STOCK	RESERVED	GOOD	(inherit)	staging
8	DEPLOYED / IN_USE	DEPLOYED	ASSIGNED	GOOD	(inherit)	with-user
9	ASSIGNED	DEPLOYED	ASSIGNED	GOOD	(inherit)	with-user
10	IN_TRANSIT / SHIPPING	DEPLOYED	IN_TRANSIT	GOOD	(inherit)	between locations
11	LOANED / ON_LOAN	DEPLOYED	ASSIGNED	GOOD	LOANER	with-user
12	LEASED	DEPLOYED	ASSIGNED	GOOD	LEASED	with-user
13	UNDER_REPAIR / IN_REPAIR	IN_MAINTENANCE	IN_REPAIR	POOR	(inherit)	service center
14	IN_MAINTENANCE	IN_MAINTENANCE	IN_REPAIR	FAIR	(inherit)	service center
15	DAMAGED	IN_STOCK / IN_MAINTENANCE	IN_REPAIR	DAMAGED	(inherit)	warehouse
16	LOST / MISSING / STOLEN	DEPLOYED	LOST	(last known)	(inherit)	last known

#	Old single status	LifecycleState	AvailabilityState	ConditionGrade	OwnershipType	Location note
17	RETIRED / DECOMMISSIONED	RETIRED	AVAILABLE	END_OF_LIFE	(inherit)	warehouse/disposal
18	DISPOSED / SCRAPPED / SOLD	DISPOSED	AVAILABLE	END_OF_LIFE	(inherit)	—

Migration: a one-time backfill script maps each legacy row via the table above, writes the four enum columns, seeds a synthetic `AssetLifecycleEvent` (`reason = 'legacy-migration'`) and an initial `ConditionLog`. "(inherit)" = carry the existing ownership/condition if already tracked elsewhere, else default OWNED/GOOD.

PART 4

Asset Detail — 7-Tab Hardware / Health Page

PART B — Asset Details / Hardware Information Redesign

The redesigned asset-detail page replaces the thin manufacturer/model/serial/OS/CPU/RAM/disk view with a **tabbed, source-attributed profile**. Every field carries a **source badge** (Manual · Agent · Intune · Discovery · Entra/Graph) and a **last-synced** timestamp, so users know provenance and freshness.

B.0 Data Population Model



Source-precedence & merge rules (per field): the reconciliation engine applies a **priority order** and **staleness window** per attribute: - **Authoritative-source-wins**: security/compliance fields (BitLocker, Defender, TPM, activation) prefer **Intune/Agent** over manual; identity/license fields prefer **Entra/Graph**. - **Manual override lock**: procurement fields (asset tag, purchase date, warranty, owner, location) prefer **Manual** unless empty, then discovery fills. - **Freshest-wins** within same tier; a field older than its staleness window is flagged "stale" in the UI. - Every write records **source**, **collected_at**, and **confidence**, enabling the source badges.

Entity mapping: Sections 1, 3 (system + hardware) map to **HardwareProfile**; Section 2 (OS) maps to **OperatingSystemInfo**; Section 2's security fields + parts of Sections 3/5 map to **SecurityPosture**; Section 4 → **InstalledSoftware**; Section 5 → **M365Profile**; Section 6 → **WarrantyContract**; Section 7 → computed **AssetHealth**.

B.1 Tab: System Information → HardwareProfile

Field	Source	Notes
Manufacturer	Agent / Discovery / Manual	CIM Win32_ComputerSystem
Model	Agent / Discovery	
Serial Number	Agent / Manual	BIOS serial; primary correlation key
Asset Tag	Manual	Procurement/CMDB
Hostname	Agent / Discovery	
Device Name (friendly)	Intune / Manual	
BIOS Version	Agent	Win32_BIOS

Field	Source	Notes
BIOS Date	Agent	
UEFI Mode	Agent	UEFI vs Legacy BIOS
Secure Boot	Agent / Intune	Enabled/Disabled
TPM Present / Version	Agent / Intune	1.2 / 2.0
Motherboard	Agent	Win32_BaseBoard
Chassis Type	Agent	Laptop/Desktop/VM/Tablet
Domain / Workgroup	Agent	
Azure AD / Entra Joined	Intune / Entra	Joined / Hybrid / Registered
Intune Enrolled	Intune	MDM enrollment state
Autopilot Registered	Intune	Autopilot device
Location / Site	Manual / Discovery	Subnet-inferred fallback
Assigned User	Manual / Intune / Entra	Primary user
Department / Office	Manual / Entra	
Purchase Date	Manual	
Warranty (summary)	Manual / Vendor API	See §B.6
Age	Derived	now - purchase_date
Lifecycle Stage	Derived / Manual	Ordered/In-stock/Deployed/Repair/Retired
Availability	Derived	In-use / Available / Reserved
Condition	Manual	New/Good/Fair/Damaged
Owner (cost center)	Manual	

B.2 Tab: Operating System → OperatingSystemInfo (+ SecurityPosture)

Field	Source	Notes
OS Name	Agent / Intune	e.g. Windows 11
Edition	Agent	Pro/Enterprise/Education
Version	Agent	e.g. 24H2
Build Number	Agent	e.g. 26100.x
Architecture	Agent	x64 / ARM64
Install Date	Agent	
Last Boot Time	Agent	Uptime derived
Update Level / Patch Status	Agent / Intune	Latest CU; missing-patch count
Activation Status	Agent	Activated / Not activated
Windows License Key	Agent / Manual	Masked; links to License Module
License Type	Agent	OEM/Retail/Volume/Subscription
License Expiry	Intune / License Module	For subscription (E3/E5)
Windows Defender Status	Agent / Intune	→ SecurityPosture
BitLocker Status	Agent / Intune	On/Off per volume → SecurityPosture

Field	Source	Notes
BitLocker Recovery Key (status)	Intune / Entra	Escrow present? yes/no ; key itself never shown here (permissioned reveal)
Firewall Status	Agent	Per-profile
Antivirus Product	Agent	Defender or 3rd-party (from Security Center)
Local Admin Accounts	Agent	Count + list (LAPS-aware)

B.3 Tab: Hardware → HardwareProfile

Field	Source	Notes
CPU (model/cores/threads/speed)	Agent / Discovery	Win32_Processor
RAM Total	Agent	
Memory Slots (used/total)	Agent	Win32_PhysicalMemoryArray
Disk(s) (model/capacity)	Agent	
Disk Usage (free/used)	Agent	Per volume
Disk Health (SMART)	Agent	Healthy/Warning/Failing
SSD vs HDD	Agent	Media type
GPU	Agent / Discovery	
Display (panel/resolution)	Agent	
Battery Health	Agent	Design vs full-charge capacity %
Battery Cycle Count	Agent	
Power Adapter	Agent / Manual	Wattage
Network Adapters	Agent	NIC list
WiFi / Bluetooth	Agent	Present + driver
MAC Address(es)	Agent / Discovery	
IP Address(es)	Agent / Discovery	v4/v6, DHCP/static
Docking Station	Agent / Manual	
USB Devices	Agent	Connected peripherals
Monitors (external)	Agent	EDID model/serial
Printer / Scanner	Agent / Discovery	
Camera / Mic	Agent	Present/enabled

B.4 Tab: Installed Software → InstalledSoftware

Field	Source	Notes
Software Name	Agent / Intune	Registry uninstall + MDM inventory
Version	Agent	
Install Date	Agent	
Publisher	Agent	
License Link	Reconciliation	FK → SoftwareLicense (Part A); shows entitlement/compliance state

Common apps surfaced with quick-status: Microsoft 365 / Office, Windows OS, Adobe Acrobat/Creative Cloud, browsers (Chrome/Edge/Firefox), Teams/Zoom, antivirus/EDR, remote tools (TeamViewer/AnyDesk), Visual Studio, AutoCAD, 7-Zip/utilities. Each installed entry is **reconciled to an entitlement**:  licensed /  unlicensed /  unmanaged.

B.5 Tab: Microsoft 365 Information → **M365Profile** (via Entra/Graph)

Field	Source	Notes
Tenant	Entra	Tenant name/ID
Assigned M365 License	Graph	assignedLicenses → links to Part A
Exchange Online	Graph	Enabled + plan
OneDrive	Graph	Provisioned + storage
SharePoint	Graph	Access/sites
Teams	Graph	Enabled
Defender for O365	Graph / Intune	Plan level
Conditional Access	Graph	Applied policies
MFA Status	Graph	Enabled/enforced/registered methods
Mailbox Size	Graph	Used/quota
Last Sign-In	Graph signIns	Feeds unused-license detection (Part A §A.8)
User Status	Entra	Active / Disabled / Blocked

B.6 Tab: Warranty & Contracts → **WarrantyContract**

Field	Source	Notes
Vendor	Manual / Vendor API	Dell/HP/Lenovo warranty lookup
Purchase Date	Manual	
Warranty Start	Manual / Vendor API	
Warranty End	Manual / Vendor API	
Remaining Days	Derived	Drives health + alerts
AMC (Annual Maintenance Contract)	Manual	Provider, coverage
Support Contact	Manual	Phone/email/portal
Invoice	Manual	Linked document
Contract Documents	Manual	Uploaded attachments

B.7 Tab: Asset Health → **computed AssetHealth**

An **overall health score (0–100)** rolled up from weighted sub-scores. Each sub-score is 0–100; the overall is a weighted average, and any sub-score in a critical band caps the overall grade.

Sub-score model

Sub-score	Weight	Inputs	Scoring logic (examples)
Battery	15%	Health %, cycle count	$\geq 80\%$ & < 500 cycles = 100; 60–79% = 70; $< 50\%$ or > 1000 cycles = 30
Storage	20%	SMART status, free space %	SMART healthy & $> 20\%$ free = 100; $< 10\%$ free = 50; SMART warning = 40; failing = 0
Memory	10%	RAM vs role baseline, slots free	Meets baseline = 100; below baseline = 50
Warranty	15%	Remaining days	$> 180d$ = 100; 90–180 = 75; 30–89 = 50; < 30 = 25; expired = 0
Security	25%	BitLocker, Defender, firewall, TPM, activation, local-admin count	All enabled = 100; each disabled control –20; unencrypted disk caps at 40
Updates	15%	Missing patches, OS support status	Fully patched = 100; each missing critical –10; unsupported OS caps at 30

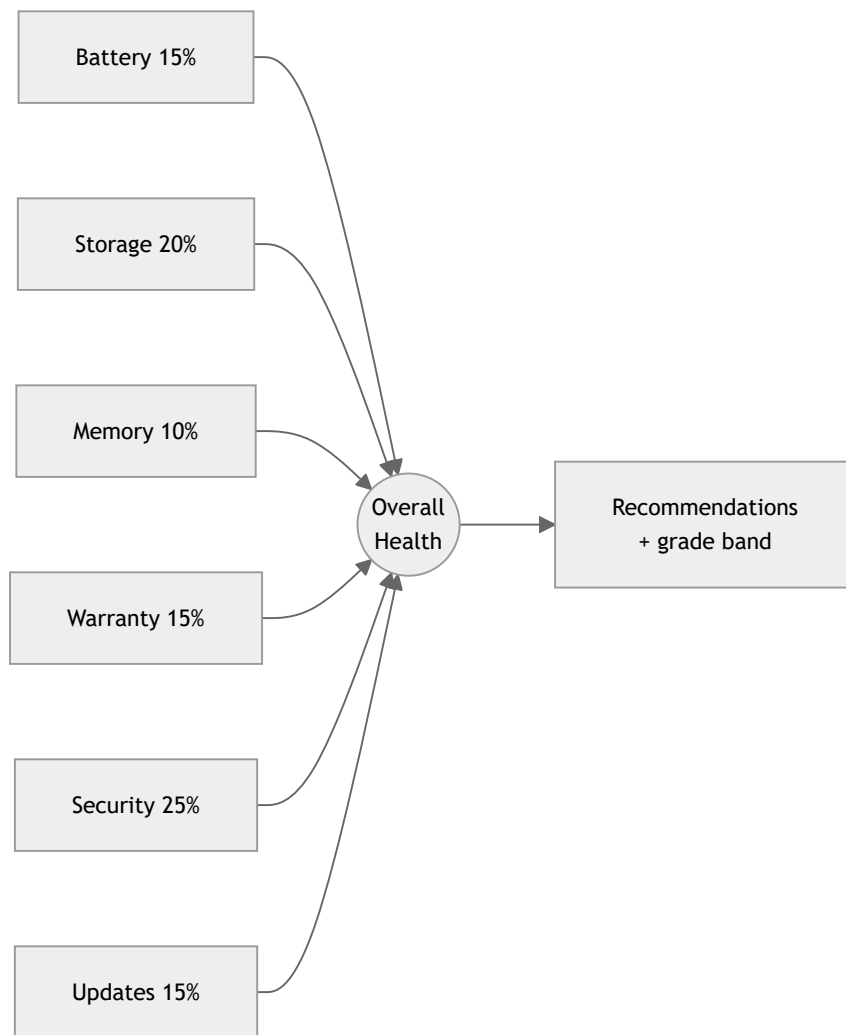
Overall grade bands: 90–100 Excellent · 75–89 Good · 60–74 Fair · 40–59 Poor · < 40 Critical. **Capping rule:** if Security or Storage < 40 (e.g. unencrypted disk or failing SMART), overall grade cannot exceed "Poor" regardless of weighted average — safety-critical dimensions dominate.

```
overall = round(  $\Sigma(\text{subScore}_i \times \text{weight}_i)$  )
if security < 40 or storage < 40:
    overall = min(overall, 59) # cannot exceed "Poor"
```

Recommendations engine

Generates prioritized, actionable items from the sub-scores, e.g.: - Storage < 50 & free $< 10\%$ → "Disk nearly full and SMART warning — schedule replacement / free space." - Battery < 50 → "Battery degraded (health 42%, 1,120 cycles) — replace battery." - Security: BitLocker off → "Enable BitLocker; drive currently unencrypted (compliance risk)." - Warranty $< 30d$ → "Warranty expires in 21 days — renew AMC or plan refresh." - Updates: unsupported OS → "Windows build past end-of-support — schedule upgrade."

Health widget



B.8 Entity Mapping Summary

Tab / Section	Primary Entity	Secondary
1 System Information	HardwareProfile	—
2 Operating System	OperatingSystemInfo	SecurityPosture
3 Hardware	HardwareProfile	—
4 Installed Software	InstalledSoftware	SoftwareLicense (Part A)
5 Microsoft 365	M365Profile	SoftwareLicense
6 Warranty & Contracts	WarrantyContract	—
7 Asset Health	AssetHealth (computed)	HardwareProfile · OperatingSystemInfo · SecurityPosture · WarrantyContract

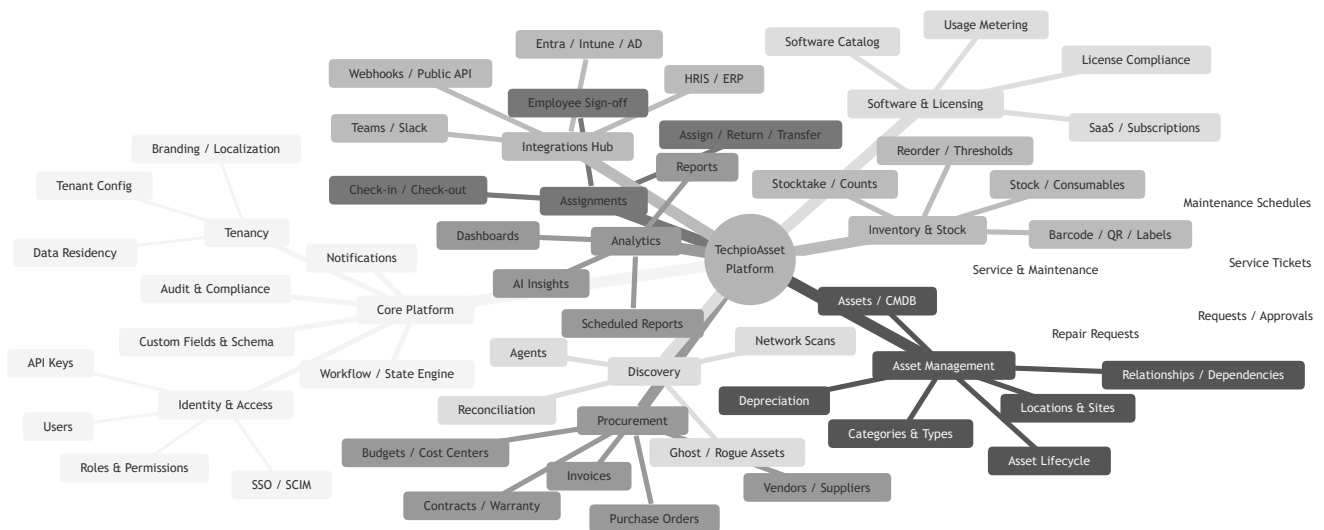
Each entity stores per-field `source`, `collected_at`, and `confidence` so the redesigned UI can render **source badges, freshness indicators, and "stale data" warnings** consistently across all seven tabs — the hallmark that lifts TechpioAsset's asset-detail experience to ServiceNow/Intune class.

PART 5

Information Architecture — Module, Screen & Navigation Trees (Web)

PART 2 — MODULE, SCREEN & NAVIGATION TREES (WEB)

2.1 Redesigned Module Map



2.2 Full Module Tree (indented)

```

TechpioAsset/
├── core/
│   ├── identity/           (auth, MFA, sessions, SSO, SCIM, API keys)
│   ├── tenancy/            (tenant provisioning, config, branding, residency)
│   ├── rbac/               (roles, permissions, ABAC policies, custom roles)
│   ├── audit/              (append-only events, security console, exports)
│   ├── notifications/      (channels, templates, digests, preferences)
│   ├── custom-fields/      (per-entity schema builder, validation)
│   ├── workflow/           (state-machine engine, approvals, guards)
│   └── files/              (upload, AV scan, object storage, thumbnails)
├── assets/
│   ├── cmdb/               (asset records, statuses, tags)
│   ├── categories/         (types, models, manufacturers)
│   ├── locations/          (sites, buildings, rooms, geo)
│   ├── lifecycle/          (states, history, disposal certificates)
│   ├── depreciation/       (methods, schedules, book value)
│   └── relationships/       (dependency graph, parent/child, CI links)
├── inventory/
│   ├── stock/              (consumables, quantities, batches)
│   ├── stocktake/          (count sessions, variance, reconciliation)
│   ├── replenishment/      (reorder points, POs trigger)
│   └── labeling/            (QR/barcode generation, print templates)
├── assignments/
│   ├── assign-return/      (checkout, checkin, transfer)
│   ├── signoff/            (employee e-signature, acknowledgment)
│   └── custody/             (chain of custody, responsibility)
├── procurement/
│   ├── vendors/            (suppliers, contacts, ratings)
│   ├── purchase-orders/    (PO lifecycle, receiving)
│   ├── invoices/           (capture, OCR, matching)
│   ├── contracts/          (warranty, SLA, renewals, expiry)
│   └── budgets/             (cost centers, spend tracking)
├── software/
│   ├── catalog/            (software titles, versions)
│   ├── licenses/           (entitlements, compliance, true-up)
│   ├── saas/               (subscriptions, seats)
│   └── metering/            (usage, reclaim unused)
├── service/
│   ├── maintenance/        (preventive schedules, tasks)
│   ├── repairs/            (repair tickets, vendor RMA)
│   └── requests/            (asset requests, approvals workflow)
├── discovery/
│   ├── agents/             (agent registry, health, tokens)
│   ├── scans/              (ingest, raw records)
│   ├── reconciliation/      (matching, merge, review queue)
│   └── anomalies/           (ghost/rogue detection)
├── integrations/
│   ├── connectors/         (Entra, Intune, AD, HRIS, ERP, Teams/Slack)
│   ├── webhooks/           (subscriptions, delivery log)
│   └── public-api/          (keys, scopes, usage)
├── analytics/
│   ├── dashboards/         (role-based, widgets)
│   ├── reports/            (builder, scheduled, exports)
│   └── ai-insights/         (health, anomalies, forecasts)
└── admin/
    ├── settings/           (system, security policies)
    ├── billing/            (plan, seats, usage)
    └── jobs/               (queue monitor, DLQ, scheduled jobs)

```

2.3 Web Screen Tree

Web App/

- └ Auth
 - └ Login (password / SSO)
 - └ MFA Challenge (TOTP/WebAuthn/Push)
 - └ Forgot / Reset Password
 - └ Accept Invite / SSO First-login
- └ Dashboard (role-based)
 - └ Admin Dashboard
 - └ Asset Manager Dashboard
 - └ IT Ops Dashboard
 - └ Finance Dashboard
- └ Assets
 - └ Asset List (filters, saved views, bulk actions)
 - └ Asset Detail (overview · history · relationships · docs · maintenance · financials)
 - └ Asset Create / Edit
 - └ Bulk Import (wizard + mapping + preview)
 - └ Category / Model / Manufacturer manager
- └ Inventory
 - └ Stock List
 - └ Stocktake Sessions
 - └ Count Session Detail (variance)
 - └ Label / QR Print Studio
- └ Assignments
 - └ Assign / Checkout Wizard
 - └ Return / Checkin
 - └ Transfer
 - └ Sign-off / Acknowledgment view
- └ Procurement
 - └ Vendors
 - └ Purchase Orders (list · detail · receive)
 - └ Invoices (capture · OCR review · matching)
 - └ Contracts & Warranty (expiry calendar)
 - └ Budgets / Cost Centers
- └ Software
 - └ Software Catalog
 - └ License Compliance (over/under-licensed)
 - └ SaaS Subscriptions
 - └ Reclaim / Optimization
- └ Service
 - └ Maintenance Schedules & Calendar
 - └ Repair Tickets
 - └ Requests & Approvals inbox
- └ Discovery
 - └ Agents (registry · health)
 - └ Scan Results
 - └ Reconciliation Review Queue
 - └ Anomalies / Ghost Assets
- └ Integrations
 - └ Connectors Catalog & Config
 - └ Sync Logs
 - └ Webhooks
 - └ Public API Keys
- └ Analytics
 - └ Dashboards (builder)
 - └ Report Library & Builder
 - └ Scheduled Reports
 - └ AI Insights
- └ Admin
 - └ Users
 - └ Roles & Permissions (custom role builder)
 - └ SSO / SCIM
 - └ Custom Fields & Workflow designer
 - └ Security (audit, sessions, policies)
 - └ Notifications templates

```
├─ Tenant Settings / Branding / Localization
├─ Billing
└─ Jobs / Queue Monitor
```

2.4 Web Navigation (Sidebar) Tree

```
└─ Sidebar
  └─ Dashboard
  └─ Assets
    └─ All Assets
    └─ Categories & Models
    └─ Locations
    └─ Import
  └─ Inventory
    └─ Stock
    └─ Stocktake
    └─ Labels & QR
  └─ Assignments
    └─ Assign / Return
    └─ Transfers
    └─ Sign-offs
  └─ Procurement
    └─ Vendors
    └─ Purchase Orders
    └─ Invoices
    └─ Contracts & Warranty
    └─ Budgets
  └─ Software
    └─ Catalog
    └─ Licenses
    └─ SaaS
  └─ Service & Maintenance
    └─ Maintenance
    └─ Repairs
    └─ Requests
  └─ Discovery
    └─ Agents
    └─ Scans
    └─ Reconciliation
    └─ Anomalies
  └─ Integrations
    └─ Connectors
    └─ Webhooks
    └─ API Keys
  └─ Analytics
    └─ Dashboards
    └─ Reports
    └─ AI Insights
  └─ Admin
    └─ Users & Roles
    └─ SSO / SCIM
    └─ Custom Fields
    └─ Security & Audit
    └─ Tenant Settings
    └─ Billing
    └─ Jobs
```

(Global top bar: tenant switcher · global search/command-palette · notifications · quick-add · profile)

*Sidebar items render conditionally by permission/ABAC; collapsible groups; pinned favorites; command-palette (**Ctrl+K**) for power users.*



PART 6

Mobile App Redesign (offline-first field companion)

PART 3 — MOBILE APP REDESIGN (Mobile-First)

Principle: the mobile app is a **purpose-built field companion**, not a shrunk web app. It optimizes the *physical* work — scanning, assigning at the desk, counting in the stockroom, capturing an invoice at receiving, signing on the device — and works **offline-first**. Web = management & analytics; Mobile = action in the field.

3.1 Mobile Navigation Tree

```

Mobile App/
├─ [Auth]
│  └─ Biometric Login (Face/Touch → device key)
│  └─ PIN fallback
│  └─ SSO (embedded browser / OIDC)
├─ ◊ Home / Dashboard (role-based)
│   └─ Quick Actions row · KPIs · My Tasks · Alerts
├─ Bottom Tab Bar (primary):
│  └─ ◊ Home
│  └─ 📷 Scan ← center FAB, always one tap
│  └─ 📋 Assets
│  └─ 📦 Inventory
│  └─ ≡ More
├─ 📷 Scan (hub)
│  └─ QR / Barcode scanner
│  └─ Batch scan mode
│  └─ Scan → Asset lookup
│  └─ Scan → Quick action (assign/return/count)
├─ 📋 Assets
│  └─ Search / Filter (voice search)
│  └─ Asset Detail
│     └─ Info · Photos · History · Docs
│     └─ Assign / Return / Transfer
│     └─ Report Issue / Maintenance
│     └─ Geotag / View on map
│  └─ Add Asset (photo + AI suggest)
├─ 📦 Inventory
│  └─ Stock Lookup
│  └─ Count Session (offline)
│     └─ Scan-to-count
│     └─ Variance view
│     └─ Submit / Sync
│  └─ Consume / Restock
├─ ⇄ Assignments
│  └─ Assign (scan → pick employee → e-sign)
│  └─ Return (scan → condition + photo)
│  └─ Transfer (scan → new holder → accept)
├─ ⚙ Tasks
│  └─ My Maintenance
│  └─ My Requests / Approvals
│  └─ Assigned To Me
├─ ≡ More
│  └─ Invoice / Receipt Capture
│  └─ Notifications
│  └─ Offline Queue & Sync status
│  └─ Profile & Devices
│  └─ Settings (Dark mode · Biometric · Language)
│  └─ Help / About

```

3.2 Role-Based Mobile Dashboards

Role	Home focus	Quick Actions
Field Technician / IT Ops	Today's maintenance tasks, assigned repairs, nearby assets	Scan, Assign, Report Issue, Start Count

Role	Home focus	Quick Actions
Asset Manager	Pending sign-offs, overdue returns, low-stock alerts, exceptions	Scan, Transfer, Approve, New Asset
Stockroom / Warehouse	Open count sessions, reorder alerts, receiving queue	Scan-to-count, Restock, Capture Invoice
Employee (self-service)	My assigned assets, request new, return, acknowledge sign-offs	Request Asset, Report Issue, Sign
Approver / Manager	Approval inbox (requests, transfers, disposals)	Approve/Reject, View asset

Each dashboard: KPI tiles (offline-cached), a **My Tasks** list, an **Alerts** feed, and a persistent **Scan FAB**.

3.3 Mobile Feature Coverage

Feature	Mobile behavior
Offline mode	Full read of scoped assets/locations + all field actions (assign/return/count/photo/sign) work offline; SQLite store; op-queue; visible sync badge.
QR + Barcode scanner	Camera scanner for QR, Code-128, EAN, DataMatrix; batch mode; torch; haptic + audio confirm; scan-then-action.
Asset lookup	Scan or search (text/voice) → instant detail even offline.
Assign / Return / Transfer	Guided flows: scan asset → pick person/location → capture condition/photo → sign → confirm. State-machine enforced.
Employee e-signature	On-device signature pad; captures name, timestamp, geolocation, asset snapshot; stored as signed acknowledgment; syncs on reconnect.
Photo upload	Attach condition photos on assign/return/issue; compressed; resumable upload; queued offline.
Invoice / receipt capture	Camera → AI OCR extracts vendor/amount/serials/warranty → user confirms → creates/links PO/invoice/asset.
Maintenance	View scheduled/assigned maintenance, check off tasks, add notes/photos, close ticket.
Inventory count	Offline count session; scan-to-increment; variance vs. expected; submit → server reconciles (additive/CRDT-friendly).
Push notifications	Approvals, assignments, overdue returns, low stock, maintenance due; deep-link into the relevant screen; actionable (approve/reject inline).
Dark mode	System-driven + manual; full theme parity (WCAG-AA palette from baseline).
Biometric login	Face/Touch ID unlock bound to secure-enclave device key; PIN fallback; step-up biometric for sensitive ops (disposal).
Voice search	Speech-to-text asset/inventory search ("find laptop tag 4821").
GPS / geotag	Capture location on assign/return/count/scan; view asset last-seen on map; geofenced site auto-detect.
Camera	Unified capture service feeding scanner, photos, invoice OCR, AI asset recognition.
Offline sync + conflict handling	Idempotent replay by opId ; optimistic baseVersion ; field-level LWW; user-prompt on hard conflict; per-op sync status UI.
Quick actions	Home-screen shortcuts + in-app FAB: Scan, Assign, Count, Capture Invoice, Report Issue.

3.4 Key Mobile Flows

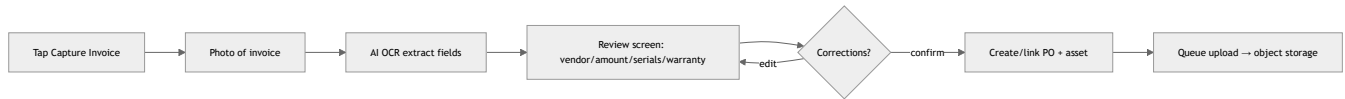
A. Scan-to-Assign (offline-capable)



B. Offline Inventory Count



C. Invoice Capture with AI



Screen highlights - Home: greeting, role KPIs, My Tasks, Alerts, prominent **Scan FAB**; offline banner + sync badge when queued ops exist. - **Scanner:** full-bleed camera, reticle, torch, batch toggle; on hit → bottom-sheet with asset summary + contextual quick actions. - **Asset Detail:** hero photo carousel, status chip, key specs, tabbed Info/History/Docs, action bar (Assign/Return/Transfer/Report), "last seen" map. - **Sign-off:** signature canvas, employee + asset summary, GPS/time stamp, submit → generates signed acknowledgment PDF on server. - **Sync Center:** list of pending ops with status (queued/syncing/conflict), conflict-resolution prompt, manual "sync now."

Cross-Cutting Non-Functional Targets

NFR	Target
Availability	99.9% (enterprise tier 99.95%)
API latency	p95 < 300 ms interactive reads; p99 < 800 ms
Scale	10k+ tenants shared-schema; millions of assets; discovery bursts of 100k+ records
RPO / RTO	≤ 5 min / ≤ 1 h
Security	SOC 2 / ISO 27001-ready; encryption at rest + in transit; RLS-enforced isolation
Accessibility	WCAG 2.1 AA (web + mobile)
Offline	Full field-action capability with idempotent, conflict-aware sync
Observability	End-to-end OpenTelemetry trace correlation request→queue→worker→DB

This blueprint keeps every existing TechpioAsset capability (QR scan, offline SQLite stocktake, biometric, provider abstractions, JWT+rotating-refresh, RBAC/scopes, RFC-9457, response envelope) and re-architects them for multi-tenant, world-class ITAM scale: a modular monolith with an outbox/event backbone, RLS-anchored hybrid tenancy, zero-trust security, SSO/SCIM/MFA, a governed human-in-the-loop AI layer, a full integrations hub, and a purpose-built offline-first mobile field app.

PART 7

Role-Based Dashboards (all 13 roles, with layouts)

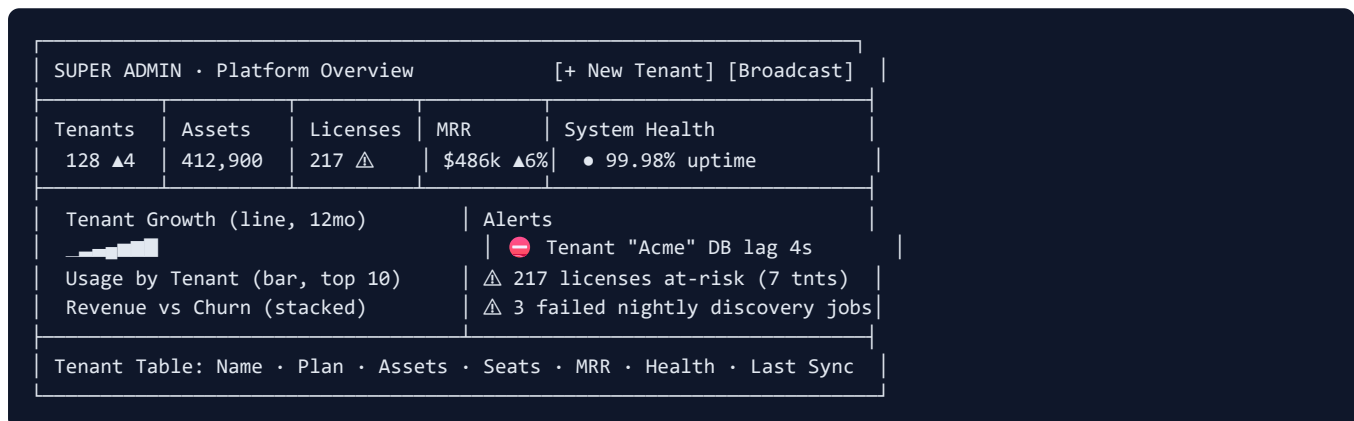
1. ROLE-BASED DASHBOARDS

Design principle: every dashboard opens with a **KPI strip (3–5 tiles)** → **primary work queue** → **charts/trends** → **alerts rail** (right drawer). Widgets are role-scoped and tenant-scoped. Each role lands on a "what needs me now" view, not vanity metrics.

Global widget grammar: - **KPI Tile** — big number + delta vs prior period + sparkline + click-through - **Work Queue** — actionable list with inline row actions - **Chart** — donut / bar / line / stacked / heatmap - **Alert Card** — severity-coded, dismissible, deep-linked - **Quick Action** — primary button cluster top-right of page

1.1 Super Admin (Platform / MSP Operator)

Cross-tenant control plane. Sees the whole estate, not one company.



Widget	Type	Data / KPI	Action
Active Tenants	KPI	Count + net new this month	→ Tenant list
Total Assets (platform)	KPI	Aggregate across tenants	→ Global asset explorer
Licenses at-risk	KPI	Over-deployed / expiring seats	→ At-risk report
MRR / Usage	KPI	Recurring revenue + delta	→ Billing
System Alerts	Alert list	Uptime, job failures, DB lag	→ Ops console
Tenant Growth	Line	New tenants / 12mo	Filter range
Usage by Tenant	Bar	API calls, storage, seats	Drill tenant
Revenue vs Churn	Stacked bar	New vs churned MRR	→ Cohorts
Tenant Health Grid	Data grid	Per-tenant status, last sync	Impersonate / suspend
Quick Actions	Buttons	New tenant, broadcast, feature-flag	—

1.2 Company Admin (Tenant Owner)

Full control of one tenant: users, roles, org structure, policies, branding.

COMPANY ADMIN · Acme Corp				[Invite User] [Settings]
Users 1,240	Assets 3,880	Utilise% 92%	Open Req 47	Compliance • 96% policy-passing
Asset Value by Category (donut)		Alerts		
User Activity (line)		⚠ 12 users with no manager set		
Requests by Status (stacked)		⚠ 3 roles over-privileged (SoD)		

Widget	Type	KPI	Action
Active Users	KPI	Licensed / active	→ User admin
Total Assets	KPI	Owned + leased	→ Asset list
Utilisation	KPI	Assigned vs stock	→ Utilisation report
Open Requests	KPI	Cross-department	→ Request inbox
Compliance Score	Gauge	Policy pass rate	→ Compliance
Asset Value by Category	Donut	Book value split	Drill category
User Activity	Line	Logins / actions	—
Role/SoD Alerts	Alert list	Over-privileged, orphan accounts	→ RBAC

1.3 IT Manager

Operational owner of the asset lifecycle. This is the "cockpit" dashboard.

IT MANAGER · Operations				[Assign] [New PO] [Report]
Pending Assignments 23	Open Maintenance 14	Warranty Expiring 30d 31 ⚠	License Utilisation 88%	Stock On-hand 126 (7 low⚠)
Assignments Queue (grid) Emp · Asset req · Dept · Age · [Do]		Alerts		
Maintenance SLA (bar by age)		⚠ 7 SKUs below reorder point		
License Utilisation (horizontal bar)		⚠ 31 warranties expire <30d		
		🚫 2 critical repairs breached SLA		

Widget	Type	KPI	Action
Pending Assignments	Work queue	Count + aging	Assign / reject
Open Maintenance	Work queue	By SLA state	→ Ticket
Warranty Expiring	KPI + list	Next 30/60/90d	→ Renew workflow
License Utilisation	Bar	% consumed per product	→ License mgr
Stock On-hand	KPI	Total + low-stock flag	→ Inventory
Maintenance SLA	Bar	Age buckets, breaches	Filter
Assignment Queue	Data grid	Requests to action	Bulk assign
Alerts	Alert rail	Reorder, SLA, warranty	Deep-link

1.4 IT Technician

Task-first, mobile-friendly. "My work today."

TECHNICIAN · My Day

[Scan QR] [Log Time]

My Tickets

Repairs

Today Tasks

SLA at risk

8

5

6

2 ⚠

My Ticket Queue (list, priority-sorted)

#4821 Laptop won't boot · P1 · SLA 2h ■■■ · [Start]

#4820 Replace RAM · P3 · [Start]

Today's Schedule (timeline)

Assigned Repairs (cards + parts)

Widget	Type	KPI	Action
My Tickets	Work queue	Open assigned	Start / resolve
Assigned Repairs	Cards	In-progress repairs + parts	Update status
Today's Tasks	Timeline	Scheduled work	Check off
SLA at risk	KPI	Breaching soon	Prioritise
Quick: Scan QR	Action	Asset lookup	Open asset
Quick: Log Time	Action	Time entry	Save

1.5 HR Manager

People-lifecycle lens: onboarding/offboarding drives asset kits.

HR MANAGER · People Ops

[New Onboarding] [Offboard]

Onboarding

Offboarding

Pending Kits

Assets to Recover

12 ▲

5

9 ⚠

14 (3 overdue⚠)

Onboarding Pipeline (kanban: Requested→Kit→Ready→Delivered)

Offboarding Pipeline (kanban: Notice→Recover→Wipe→Closed)

Kit Readiness (bar by start-date)

Alerts: 3 kits missing laptop

Widget	Type	KPI	Action
Onboarding count	KPI	Upcoming starts	→ Pipeline
Offboarding count	KPI	Upcoming exits	→ Pipeline
Pending Kits	Work queue	Kits not ready	Expedite
Assets to Recover	Work queue	Outstanding returns	Chase / escalate
Onboarding Pipeline	Kanban	Stage counts	Advance stage
Offboarding Pipeline	Kanban	Recovery/wipe status	Advance stage
Kit Readiness	Bar	Ready vs start date	—

1.6 Finance Manager

Money view of the asset base: capex, depreciation, license cost, invoice control.

FINANCE · Asset Financials					[Export]	[Close Period]
Invoices to Verify: 18	Budget vs Spend: 74%	Depreciation MTD: \$92k	License Cost/mo:\$61k	Book Value \$4.2M		
Budget vs Spend (bar by dept)		Alerts				
Depreciation Schedule (line)		⚠ 3 depts over budget				
License Cost Trend (area)		⚠ 18 invoices unmatched to PO				
Invoice Verification Queue (grid)		🚫 2 duplicate invoices flagged				

Widget	Type	KPI	Action
Invoices to Verify	Work queue	3-way match pending	Verify / dispute
Budget vs Spend	Bar	% consumed by dept	Drill dept
Depreciation	Line/KPI	MTD + method	→ Schedule
License Cost	Area	\$/mo trend	→ License spend
Book Value	KPI	Net asset value	→ Register
Duplicate/Anomaly	Alert	Flagged invoices	Investigate

1.7 Procurement Manager

Purchasing pipeline: PR → approval → PO → delivery.

PROCUREMENT · Purchasing					[New PR]	[New PO]
Open PRs 16	Open POs 22	Pending Approvals:9	Deliveries Due 7d: 11	Budget Left \$340k		
PO Board (kanban: Draft→Approval→Ordered→Partial→Received)		Alerts				
Spend by Vendor (bar)		⚠ 4 POs overdue delivery				
Cycle Time (PR→PO) (line)		⚠ 9 approvals aging >3d				

Widget	Type	KPI	Action
Open PRs	Work queue	Awaiting conversion	→ Create PO
Open POs	Kanban	By status	Advance
Pending Approvals	Work queue	In approval chain	Nudge
Deliveries Due	List	Next 7d	Receive
Budget Remaining	KPI	vs allocation	→ Budget
Spend by Vendor	Bar	Top vendors	Drill
Cycle Time	Line	PR→PO days	—

1.8 Inventory Manager

Stockroom control: levels, reservations, cycle counts.

INVENTORY · Stockroom		[Receive] [Adjust] [Count]		
SKUs	Low Stock	Reserved	Count Due	Stock Value
342	7 ▲	58	19	\$612k
Stock Level by Category (bar w/ reorder line)				
Low-Stock Watchlist (grid: SKU·On-hand·Reorder·Reserved·[Reorder])				
Movement (in/out, line)		Alerts: 7 below reorder, 2 zero		

Widget	Type	KPI	Action
Total SKUs	KPI	Active stock items	→ Catalog
Low Stock	Work queue	Below reorder point	Reorder
Reserved	KPI	Allocated not issued	→ Reservations
Count Due	Work queue	Cycle-count schedule	Start count
Stock Value	KPI	On-hand valuation	→ Report
Stock by Category	Bar+threshold	Level vs reorder	—
Movement	Line	In/out flow	—

1.9 Office Admin

Facility-level custodian; light asset + request handling for a site.

OFFICE ADMIN · HQ Site		[Log Asset] [Request]	
Site Assets	Open Reqs	Visitors/ Deliveries:4	Consumables Low
540	9		5 ▲
Site Requests Queue (list)		Consumables (bar)	
Recent Deliveries (list)		Alerts: printer toner low x2	

Widget	Type	KPI	Action
Site Assets	KPI	Assets at location	→ Site list
Open Requests	Work queue	Facility/asset reqs	Fulfil
Deliveries	List	Inbound today	Receive
Consumables Low	Alert list	Toner, cables, etc.	Reorder

1.10 Department Manager

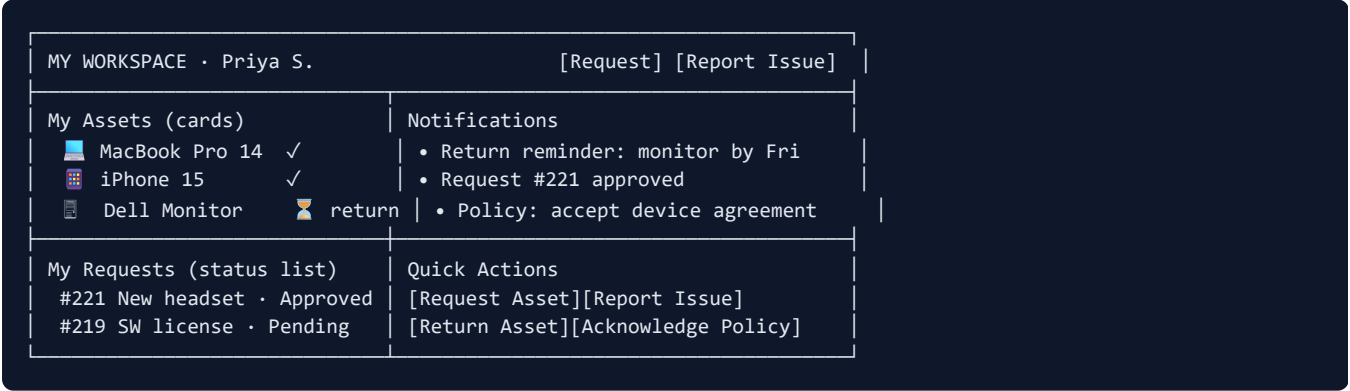
Team-scoped view: their people's assets, approvals, requests.



Widget	Type	KPI	Action
Team Assets	KPI	Assigned to team	→ Team list
Pending Approvals	Work queue	Requests to me	Approve/reject
Open Requests	Work queue	Team's submitted	Track
Budget Used	Gauge	vs team allocation	—
Team Asset Grid	Data grid	Per-employee holdings	Reassign

1.11 Employee (Self-Service)

Consumer-grade simplicity. "My stuff, my requests."



Widget	Type	KPI	Action
My Assets	Cards	Currently held	View / return / report
My Requests	Status list	Open + history	Track
Notifications	Feed	Reminders, approvals, policies	Act
Quick Actions	Buttons	Request, report, return, ack	—

1.12 Auditor (Read-mostly Compliance)

Evidence and exceptions. No mutation of assets; can annotate/flag.

AUDITOR · Compliance Posture

[Export Evidence] [Sample]

Compliance Score: 94%

Open Exceptions:23

SoD Conflicts:4

Audit Coverage 87% assets verified

Exceptions by Type (bar)

Audit Trail (immutable log stream)
who·what·when·before→after

Control Status (heatmap)
Alerts: 4 licenses over-deploy
2 assets missing since count

Widget	Type	KPI	Action
Compliance Score	Gauge	Weighted control pass	→ Controls
Open Exceptions	Work queue	Unresolved findings	Flag / note
SoD Conflicts	Alert list	Role conflicts	→ RBAC evidence
Audit Coverage	KPI	% verified in cycle	→ Sampling
Exceptions by Type	Bar	Category breakdown	Drill
Audit Trail	Log stream	Immutable change log	Filter/export
Control Heatmap	Heatmap	Domain × status	Drill

1.13 Vendor (External, Scoped Portal)

Sees only their own POs, service tickets, invoices for this tenant.

VENDOR PORTAL · Dell Technologies

[Submit Invoice] [Update]

Assigned POs 12

Service Tickets: 5

Invoices Pending: 3

On-time Delivery 96%

My POs (list: PO#·Items·Value·Due·Status·[Update ETA])

Service Tickets (list assigned to us)

Invoice Status (list)

Alerts: 2 POs need ETA update

Widget	Type	KPI	Action
Assigned POs	Work queue	Open orders	Update ETA / ship
Service Tickets	Work queue	Assigned repairs/RMA	Update status
Invoices	Status list	Submitted + paid	Submit new
On-time Delivery	KPI	SLA scorecard	—

PART 8

Reports Catalogue (what each screen must surface)

2. REPORTS CATALOGUE

Formats key: **C**=CSV · **X**=XLSX · **P**=PDF. Schedule = can be scheduled/emails. Permission uses RBAC scopes.

2.1 Asset Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Asset Register	Full inventory of record	Tag, Name, Category, Status, Owner, Location, Cost, Book Value	Category, Status, Location, Dept	C/X/P	✓	asset.read
Assignment History	Who held what, when	Asset, Employee, From, To, Duration	Asset, Employee, Date	C/X/P	✓	asset.read
Asset Utilisation	Assigned vs idle	Category, Total, Assigned, Idle, %	Category, Location	X/P	✓	asset.report
Asset Movement	Transfers between sites/people	Asset, From, To, Type, Date, By	Type, Date	C/X	✓	asset.read
End-of-Life / Disposal	Retirement & e-waste	Asset, EOL Date, Method, Value Recovered	Date, Method	C/X/P	✓	asset.dispose
Asset Aging	Age distribution & refresh due	Category, Age band, Count, Refresh due	Category	X/P	✓	asset.report

2.2 License Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
License Position	Entitled vs deployed	Product, Entitled, Deployed, Available, Compliance	Product, Vendor	X/P	✓	license.read
Over/Under-Deployment	Compliance exposure	Product, Delta, \$ Exposure	Status	X/P	✓	license.report
Renewal Forecast	Upcoming renewals	Product, Seats, Renew Date, Cost	Window	X/P	✓	license.read
Seat Assignment	Who has each seat	Product, User, Assigned, Last Used	Product, Idle	C/X	✓	license.read
Reclaimable Seats	Idle/unused seats	Product, User, Idle days, \$ Save	Idle threshold	X/P	✓	license.report

2.3 Inventory Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Stock On-hand	Current levels	SKU, Name, On-hand, Reserved, Reorder	Category, Location	C/X	✓	inventory.read
Low-Stock / Reorder	Replenishment	SKU, On-hand, Reorder pt, Suggested Qty	Location	X/P	✓	inventory.read
Stock Movement	In/out ledger	SKU, Type, Qty, Balance, Date, By	Type, Date	C/X	✓	inventory.read
Cycle Count Variance	Count accuracy	SKU, System, Counted, Variance, \$	Count batch	X/P	✓	inventory.count
Stock Valuation	On-hand value	SKU, Qty, Unit Cost, Total	Method, Location	X/P	✓	inventory.report

2.4 Finance Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Depreciation Schedule	Book value over life	Asset, Method, Cost, Accum Dep, NBV	Method, Period	X/P	✓	finance.read

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Budget vs Actual	Spend control	Dept, Budget, Spent, Variance, %	Dept, Period	X/P	✓	finance.report
Capex/Opex Split	Capital vs operating	Category, Capex, Opex, Total	Period	X/P	✓	finance.report
Total Cost of Ownership	Lifetime cost per asset	Asset, Purchase, Maint, License, TCO	Category	X/P	✓	finance.report
Invoice Reconciliation	3-way match status	Invoice, PO, GRN, Status, Variance	Status	C/X	✓	finance.verify

2.5 Warranty Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Warranty Expiry	Coverage lapse risk	Asset, Vendor, Start, End, Days left	Window, Vendor	X/P	✓	asset.read
Warranty Coverage	Covered vs uncovered	Category, Covered, Uncovered, %	Category	X/P	✓	asset.report
Claims History	Warranty claims made	Asset, Claim, Date, Outcome, \$ Recovered	Date, Outcome	C/X	✓	asset.read

2.6 Maintenance Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Ticket Summary	Volume & status	Ticket, Asset, Type, Priority, Status, Age	Status, Priority	C/X/P	✓	maint.read
SLA Compliance	Response/resolve SLA	Ticket, SLA, Target, Actual, Met?	Period, Tech	X/P	✓	maint.report
MTTR / MTBF	Reliability metrics	Category, MTTR, MTBF, Failures	Category	X/P	✓	maint.report
Maintenance Cost	Repair spend	Asset, Labor, Parts, Total	Period, Asset	X/P	✓	finance.read
Technician Load	Workload balance	Tech, Open, Closed, Avg time	Period	X	✓	maint.report

2.7 Audit / Compliance Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Audit Trail	Immutable change log	Timestamp, Actor, Entity, Before→After, IP	Entity, Actor, Date	C/P	✓	audit.read
Exception Register	Open compliance findings	Exception, Type, Severity, Owner, Age	Type, Severity	X/P	✓	audit.read
Policy Compliance	Control pass/fail	Control, Domain, Status, Evidence	Domain	X/P	✓	audit.report
Access / SoD Review	Role conflicts	User, Roles, Conflict, Risk	Risk	X/P	✓	audit.rbac
Chain of Custody	Asset custody proof	Asset, Custodians, Transfers, Signatures	Asset	P	✗	audit.read

2.8 Employee Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Assets by Employee	Holdings per person	Employee, Assets, Value, Since	Dept, Status	C/X	✓	employee.read
Onboarding Status	Kit readiness	Employee, Start, Kit, Status	Status, Date	X/P	✓	hr.read
Offboarding / Recovery	Return tracking	Employee, Exit, Assets Out, Recovered	Status	X/P	✓	hr.read
Acknowledgement Log	Policy sign-offs	Employee, Policy, Signed, Version	Policy	C/P	✓	hr.read

2.9 Vendor Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Vendor Scorecard	Performance rating	Vendor, On-time %, Quality, Disputes	Period	X/P	✓	vendor.report
Spend by Vendor	Vendor concentration	Vendor, POs, Spend, % of total	Period	X/P	✓	procure.report

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
Contract Expiry	Vendor contracts due	Vendor, Contract, End, Auto-renew	Window	X/P	✓	vendor.read
SLA Adherence	Service SLA per vendor	Vendor, Tickets, Met, Breached	Period	X/P	✓	vendor.report

2.10 Procurement Reports

Report	Purpose	Key Columns	Filters	Fmt	Sched	Permission
PO Status	Order pipeline	PO, Vendor, Value, Status, ETA	Status, Vendor	C/X	✓	procure.read
PR-to-PO Cycle Time	Process efficiency	PR, PO, Days, Approver	Period	X/P	✓	procure.report
Approval Aging	Bottleneck detection	Request, Stage, Days waiting, Approver	Stage	X	✓	procure.report
Receiving / GRN	Delivery confirmation	PO, Item, Ordered, Received, Variance	Status	C/X	✓	procure.receive
Procurement Spend	Buying analysis	Category, Vendor, Qty, Spend	Period, Category	X/P	✓	procure.report

2.11 Executive Dashboard (Report Pack)

Report	Purpose	Key Columns / KPIs	Filters	Fmt	Sched	Permission
Estate Summary	One-page asset health	Total assets, value, utilisation, growth	Period, Dept	P	✓	exec.read
Cost & Savings	Financial outcomes	TCO, license savings, reclaimed seats, budget	Period	P	✓	exec.read
Risk & Compliance	Posture snapshot	Compliance %, exceptions, warranty lapse, SLA	Period	P	✓	exec.read
Lifecycle KPIs	Operational health	MTTR, refresh due, aging, onboarding time	Period	P	✓	exec.read
Trend Board	Directional signals	Assets, spend, tickets, utilisation trends	Range	P	✓	exec.read

PART 9

UI Design System + Wireframes (tokens, components, screens)

3. UI DESIGN SYSTEM + WIREFRAMES

3.1 Visual Language

Design DNA: Microsoft Intune's dense clarity + ServiceNow's workspace shell + Azure Portal's blade/drawer model + Freshservice's friendly warmth.

Layout model - **Left nav** (collapsible, 2-level, icon+label) · **top bar** (tenant switcher, global search, command palette , notifications, help, avatar) · **content canvas** (cards + data grid) · **right drawer** (contextual detail/edit, non-navigational). - Max content width fluid; 12-column grid; 8px spacing base.

Color palette (professional, WCAG-AA ≥ 4.5:1 text)

Token	Light	Dark	Use
--bg	#F5F7FA	#0F1419	App background
--surface	#FFFFFF	#1A2027	Cards, drawer
--surface-2	#EEF1F5	#242C35	Subtle fills, grid header
--border	#E2E8F0	#2E3846	Dividers
--text	#1A2027	#E7ECF2	Primary text
--text-muted	#5A6472	#9AA6B4	Secondary
--primary	#2563EB	#4F8BFF	Brand / actions
--success	#16A34A	#34D06C	In-service, healthy
--warning	#D97706	#F5A623	At-risk, expiring
--danger	#DC2626	#F26D6D	Breach, over-deploy
--info	#0891B2	#38BDF8	Neutral status
--accent	#7C3AED	#A78BFA	Highlights, focus

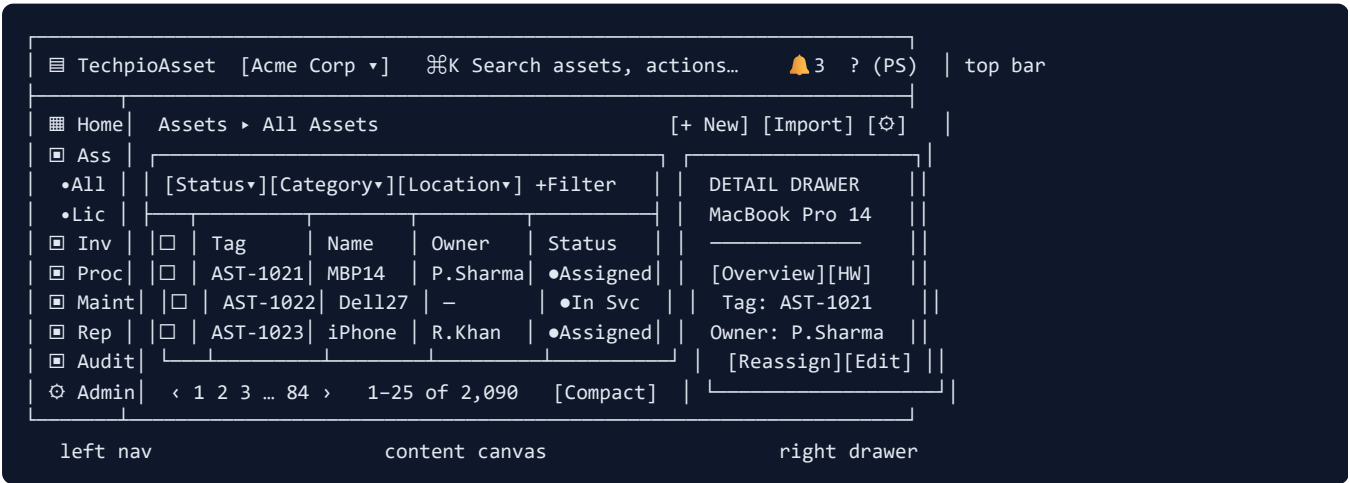
Status badges — pill shape, tinted bg + solid dot:  **In Service** (green),  **Assigned** (blue),  **In Repair** (amber),  **Retired** (grey),  **Reserved** (violet),  **Over-deployed** (red).

Components - **Cards:** 12px radius, 1px border, subtle elevation on hover; header + KPI + sparkline. - **Data grid:** sticky header, column pin/resize/reorder, density toggle (comfortable/compact), inline row actions, bulk-select toolbar, saved views, server-side paging + column filters. - **Charts:** donut, horizontal/vertical bar, line/area, stacked, heatmap, gauge — colorblind-safe series palette, always with data labels + legend + empty/loading/error states. - **Timeline:** vertical event rail for asset history/maintenance. - **Right drawer:** 480px, tabbed, edit-in-place, sticky footer actions; ESC/overlay dismiss. - **Quick filters:** chip row above grid + advanced filter builder. -

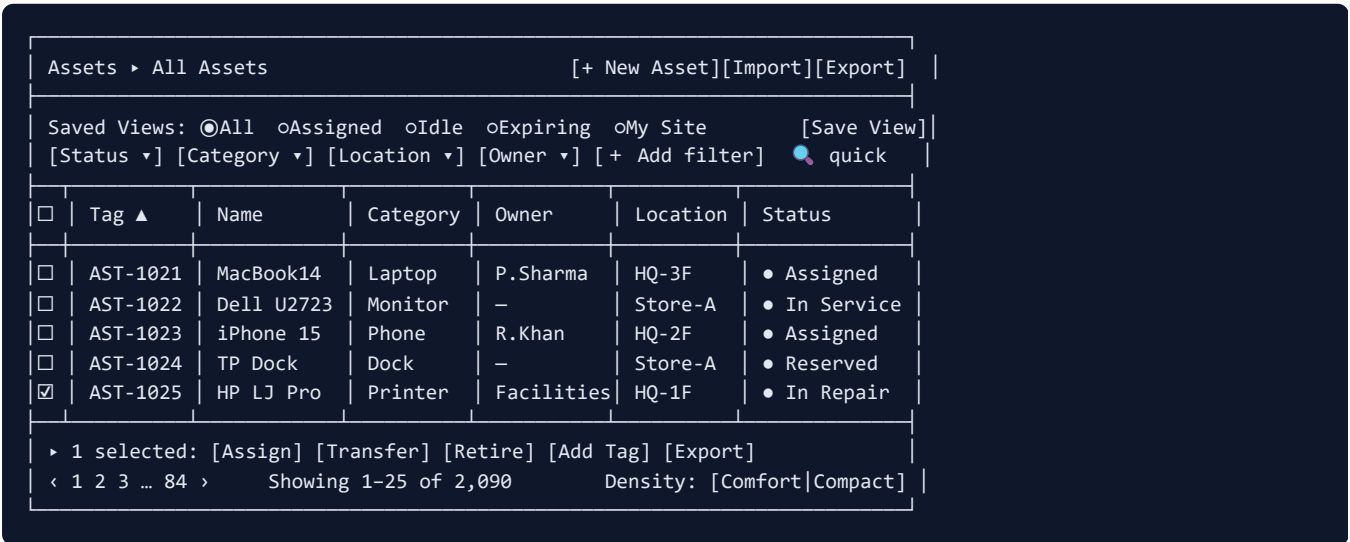
Command palette (): search assets, run actions, jump to screens. - **Typography:** Inter / Segoe UI; 14px base grid, 13px dense; tabular numerals for metrics. - **Motion:** 150–200ms ease; drawer slide, skeleton loaders; respects **prefers-reduced-motion**. - **Responsive:** left nav → hamburger < 1024px; grid → cards < 768px; right drawer →

full-screen sheet on mobile. - **A11y**: focus rings, ARIA grid roles, keyboard nav, 44px touch targets, contrast-checked tokens, not color-alone status (icon+label).

3.2 Global Layout



3.3 Asset List (Data Grid + Filters)



3.4 Asset Detail (Tabbed, Right Drawer or Full Page)



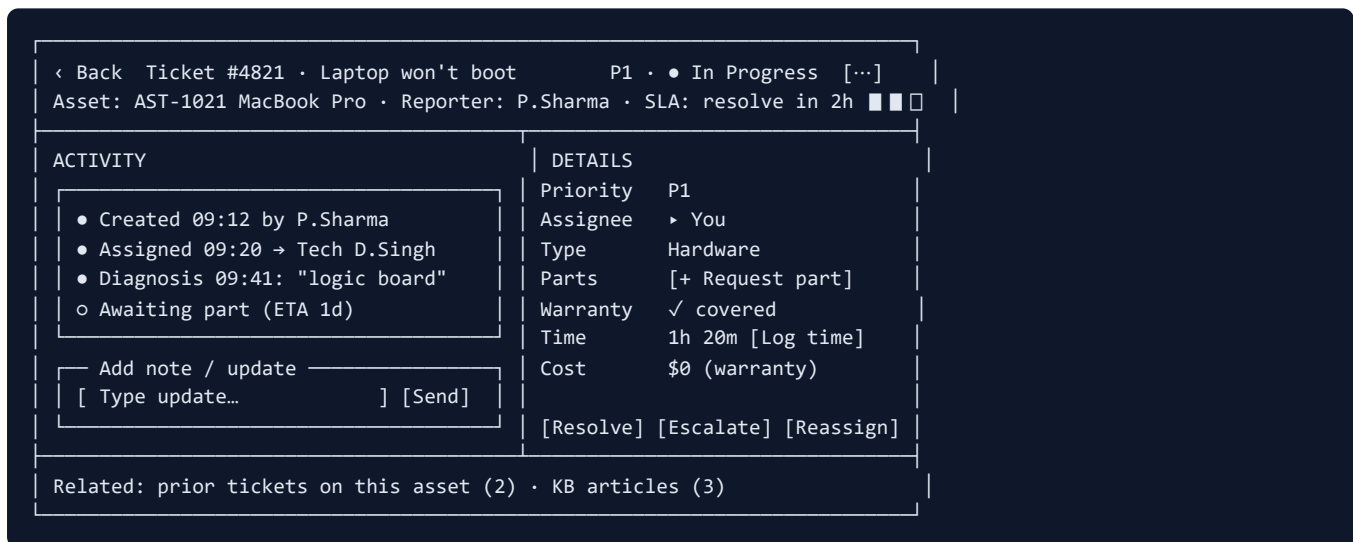
3.5 License Management (Seat-Pool Cards + Assignment Grid + Error State)



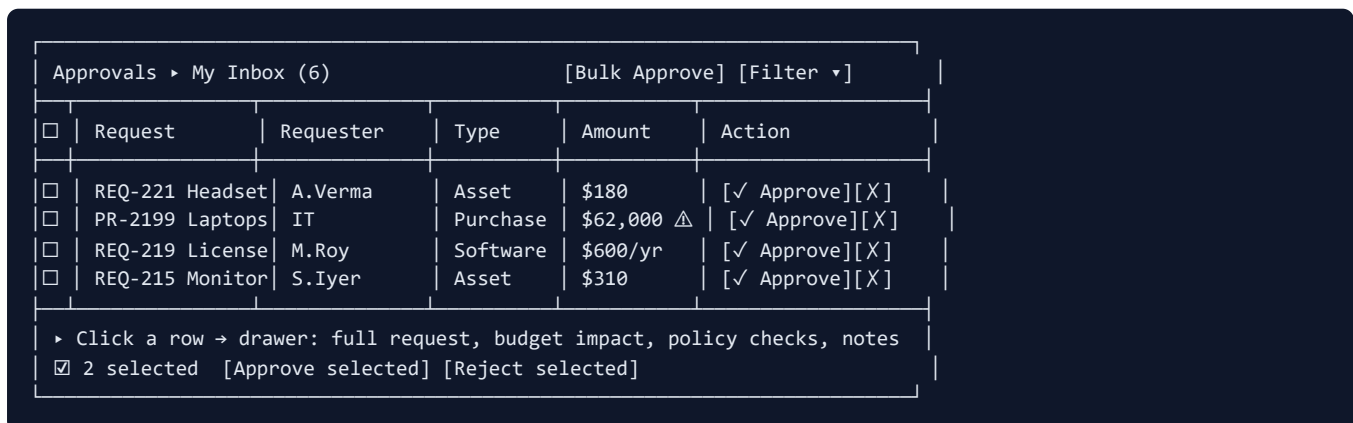
3.6 Procurement (PO Board — Kanban)



3.7 Maintenance Ticket



3.8 Approval Inbox



3.9 Mobile Screens

Home Dashboard (Employee/Tech)



QR Scan → Asset



Assign / Return Sheet

AST-1021

MacBook Pro 14

Return Asset

Condition ▾

• Good

Location ▾

Store-A

Notes...

 Add photo

 Signature

Confirm

← swipe-up bottom sheet